Irrigation UI v2 → firmware implementation plan

Oct 30, 2026 · @joe bela

Summary

The v2 UI (`portal/app-demo.html` + `mock.js`) needs 21 HTTP routes and about 15 firmware behaviours; today's firmware serves the v1 UI and covers 9 routes and roughly a third of the behaviours. The hardware layer (`Hardware.h`) is mostly ready: the gap is in the logic above it.

- **Hardware classes:** keep them. Needed changes are small: runtime calibration, a 3 s long press with a press-start event, LED/buzzer pattern players, a test mode and configurable daily cap in `WaterController` .
- **Logic layer:** the start-window scheduler must become a calendar scheduler (one-off and weekly entries per zone), plus DND windows, a run queue, three watering modes, deficit dosing, cost counters, alert edge detection and a weather cache.
- **Storage:** settings grow from ~80 bytes to several KB (calendars, plant/soil/sprinkler parameters, cost). `SettingsData` v2 plus a separate calendar store.
- **API:** rewrite `Api.h` against `API-v2.md` , using ArduinoJson for `/api/state` (the current 256-byte printf fragments will truncate UTF-8 zone names).
- **Safety rule kept:** every path that moves water (schedule, Water now, button, output test) still goes through `WaterController` .

Sources read: `portal/app-demo.html` , `portal/mock.js` , `claude/API-v2.md` , `claude/code-review.md` , `Hardware.h` , `PINS.h` , `Config.h` , `main.cpp` , `claude/Settings.h` , `claude/Scheduler.h` , `claude/Api.h` , `claude/WifiPortal.h` , the handoff note.

UI feature inventory

Every feature below is driven by data the firmware must produce or by a request it must handle. "Client only" features need nothing from the ESP except serving the page.

Screen	Feature	What the firmware must do
Global	2 s state polling (6 s when the tab is hidden), offline banner after 2 failed polls	Answer `GET /api/state` fast (< 100 ms) and never block the loop
Global	Header error/warning badges, problem list	Supply `flt`, `ok` (clock), `tank.low`, `env.ok`, per-zone `sf`, `wifi.rssi`, `g.blk`, `g.off`, `g.dnd`
Global	Clock drift warning (> 90 s vs phone)	Send accurate `t`; UI compares it to the phone
Home	Summary card: temperature, humidity, tank %, raining icon	`env.c/h`, `tank.pct` (from tankE/tankF calibration), `rain`
Home	4 zone cards: name, mode, covered, status chip (Watering, Queued, DND, Waiting + reason, Fault, All off)	Compute `st` + `why` per zone in a fixed precedence (flowchart 7.6)
Home	Soil bar with lo–hi band, "est." striped bar in open loop, low-colour below `lo`	`soil` (measured %, or water-balance estimate when m = 2), `sf`, `lo`, `hi`
Home	Last run (time, length, stop reason), Next run (time, length or "auto", predicted skip reason)	`last` from the log, `next` from the calendar for the next 8 days, with skip prediction (ALL_OFF, DND, RAIN)
Home	Enable switch with Undo toast	`POST /api/zone en`; disabling a running zone stops it
Home / Zone	Water now sheet: deficit fill, 1/2/5/10/15/30 min, until a time (≤ 60 min), stop	`POST /api/water` with result codes; `def` = deficit dose; `def: {sec,mm}` in state
Home / Zone	DND sheet: 1/4/8/24 h, until turned off, until a date/time, turn off	`POST /api/dnd`; zone and global DND, `FOREVER = 4102444800`
Home / Zone	Conflict dialogs: DND blocks Water now → "override"; DND would stop a running zone; cold limit → "water anyway"	Return `r:DND` with kind `z/g/w`, `r:COLD` with `c` and `lim`; accept `dnd=1`, `force=1`
Zone	Rename (UTF-8, emoji, 24 code points)	Store up to 96 bytes, count code points, JSON-escape

Screen	Feature	What the firmware must do
Zone	Mode: Schedule / Closed loop / Open loop, with explanation	`m` 0/1/2; switching to Schedule fills empty lengths with 900 s and answers `fill:n`
Zone	Status block: soil, last, result, next, actions	Same state fields as home
Zone	3-week calendar strip (last, this, next week) with per-day chips, "Other zones" overlay	`GET /api/zone` (`sch` , `dsch` , `gdsch`), `GET /api/calendar`
Zone	Day list: waterings and DND windows, nested when a watering falls inside DND (shown as skipped)	Store entries; the firmware must also skip them for real
Zone	Add/edit watering: start time, length (Schedule only), weekdays or one-off date	`POST /api/sched` (sid 0 = new, del = 1)
Zone	Add/edit DND window: start, end (across midnight, multi-day one-off), zone or all zones	`POST /api/dsch` (id 0 = global); an active window stops a covered run (result DND)
Zone	Plant / Soil / Sprinkler presets and 9 numeric parameters	Store `plant{p,kc,root,lo,hi}` , `soil{p,fc,wp,inf}` , `spr{p,rate,eff}` ; reject `lo >= hi`
Zone	Min. temperature stepper (frost limit … 30 °C), Covered switch	`cold` ⩾ global `frost` ; `cov` removes rain credit and rain blocking
Zone	Recent activity (5 rows), link to full history	`hist` = last 5 log rows of that zone
History	Filter chips: all, per zone, problems, cost	`GET /api/log` CSV with run, SKIP, ALERT and COST rows
Health	Problems with action buttons: Clear fault, Set clock, Turn off All off / global DND	`/api/faults/clear` , `/api/clock` , `/api/global` , `/api/dnd`
Health	Sensors list (T, RH, p, tank, rain, 4 soil) and Controller list (clock, pump, Wi-Fi, uptime, firmware, hardware)	State fields `lt` , `pump` , `up` , `ver` , `hw` , `wifi.ssid`
Health	Forecast today/tomorrow: Tmax/Tmin, ET_0, rain mm, rain chance, last-update note	`GET /api/weather` from a cached Open-Meteo fetch

Screen	Feature	What the firmware must do
Health	Output test: 4 valves (60 s, several at once), pump (follows valves, or alone 10 s), error LED 3 s, buzzer; live countdown, stop	`POST /api/test`, `tst:[{o,z,u}]` in state; refused while watering
Settings	Controller name = hostname (`A–Z a–z 0–9 -` , 1–24)	Validate, store, apply to mDNS/DHCP hostname (needs reboot or `MDNS` restart)
Settings	Frost limit, daily limit per zone, rain stop threshold (mm), queue timeout	Store and use in the rules; raising frost raises zone `cold` values below it
Settings	Weather location lat/lon	Open-Meteo request parameters; also Hargreaves fallback (latitude)
Settings	Calibration: pump flow L/min, tank empty/full distance, live distance; 4 soil sensors dry/wet with live raw value	`GET /api/settings` with `tankNow`, `soilCal[].raw`; `POST /api/cal` (dry > wet)
Settings	Button short press on/off, buzzer on/off	`btn`, `buzz` flags read by the button and feedback code
Settings	Cost: basis off/water/power, prices, pump power, currency; totals today/week/month/season/year; log periods; reset; "keep or zero" on a price change	Count pump seconds per period; roll periods at local boundaries; write COST rows; `cr=1`; `/api/cost/reset`
Settings	Clock: controller time, set from this device	`POST /api/clock t`
Settings	Wi-Fi: SSID and RSSI, change network (reboot into setup AP)	`POST /api/wifi/reset` → clear credentials or force AP, restart
Settings	Download log CSV, plain-text health page	`/log.csv`, `/health`
Menu	Global DND, All off (= winter mode) switch	`/api/dnd id=0`, `/api/global off`; switching All off off starts a new cost season
Client only	Language HU/EN, theme, colour palette, old-browser fallback, mowing game, rain easter egg, cactus warning, desktop sidebar	Nothing; only keep `/health` working for the old-browser link

API endpoints vs firmware

9 API routes exist, 5 of them in the v1 shape; 12 are missing. Status is measured against `claude/Api.h` and `WifiPortal.h`.

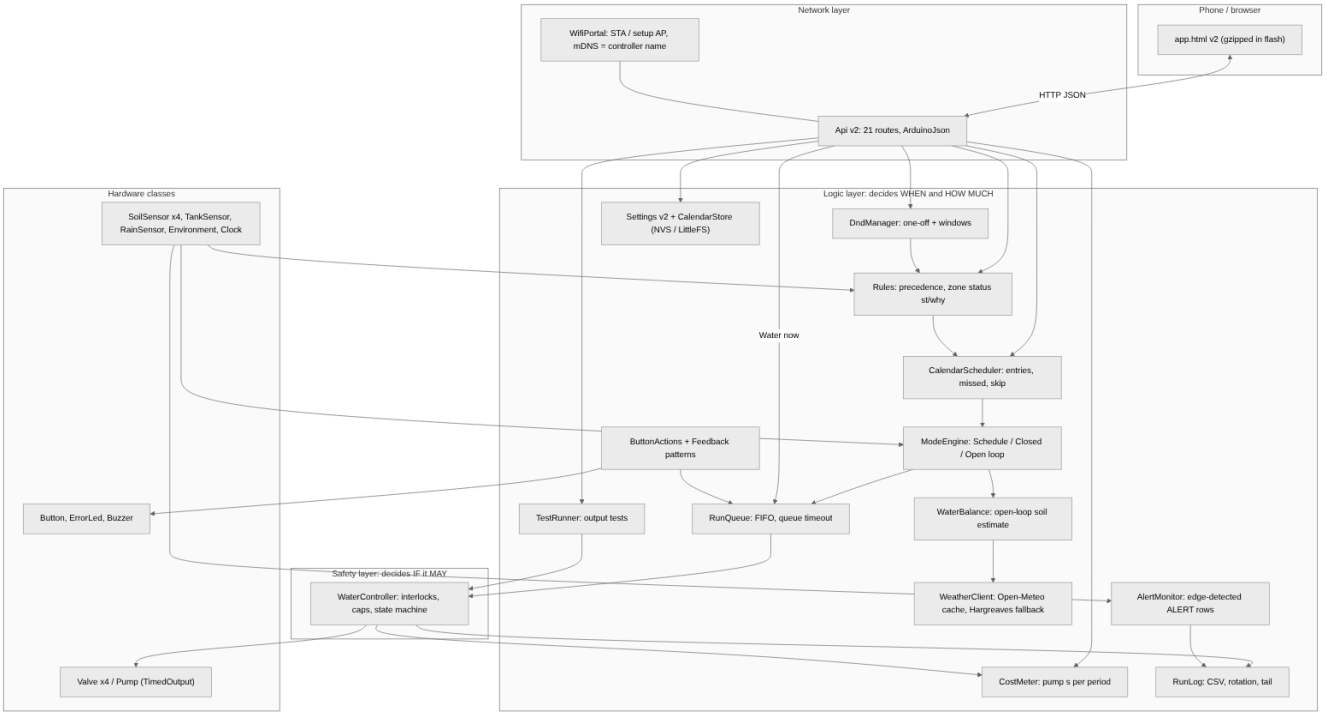
Route	Purpose	Today	Work needed
`GET /api/state`	Everything the home, zone and health views show	v1 shape	Rebuild: `name ver hw`, `tank.pct`, `g.dnd`, `tst`, zone `st/why/dnd/lo/hi/last/next/def`; drop `win`, `run/s/e`
`GET /api/zone?id=`	Zone config + calendar + last 5 log rows	Missing	New
`POST /api/zone`	name, en, m, cold, cov, plant., *soil.*, spr.*	v1 (run/s/e/mode/cov date)	Rewrite; `fill` answer; `band` and `cold` errors
`POST /api/sched`	Watering calendar entries	Missing	New
`GET /api/calendar`	All zones' calendars	Missing	New
`POST /api/dnd`	One-off DND (zone or global)	Missing	New; stops a covered run
`POST /api/dsch`	DND windows	Missing	New; stops a covered run
`POST /api/water`	Water now	v1 (no queue, no DND/COLD dialog, no def)	Rewrite to the precedence in flowchart 7.2
`POST /api/stop`	Stop and clear the queue	Exists	Also clear the queue
`POST /api/global`	All off	Exists (+ `win`, `frost`)	Drop `win`; season start on off→0
`POST /api/clock`	Set RTC	Exists	Keep; log RTC_OK
`POST /api/faults/clear`	Clear latched faults	Exists	Keep; log FAULT_OK
`POST /api/wifi/reset`	Reboot into setup AP	Missing	New (WifiPortal method)
`GET/POST /api/settings`	Global settings, totals	Missing	New
`POST /api/cal`	Soil calibration	Missing	New
`POST /api/test`	Output test	Missing	New (TestRunner)

Route	Purpose	Today	Work needed
`POST /api/cost/reset`	Reset cost counters	Missing	New
`GET /api/weather`	Cached forecast	Missing	New (WeatherClient cache)
`GET /api/log`	Log CSV	v1 columns, last 6 KB	New columns; decide size (section 9)
`GET /log.csv`	Full download	Exists	Keep
`GET /health`, `/`	Plain-text status, gzipped UI	Exists	Regenerate `AppPage.h` from the v2 page (without the mock and demo panel)

Build note: `app-demo.html` contains the mock (`window.fetch` override) and the Demo panel. The firmware page must be built from the same source with those two blocks removed, or the phone will talk to the fake controller.

Firmware architecture

Keep the existing three-layer rule and add the missing middle: requests flow down, only `WaterController` touches the pump and valves, and everything runs in the single Arduino loop except the HTTPS weather fetch.



Layer	Rule
Hardware	One class per device, non-blocking `update()`, fail-safe defaults (tank starts LOW, outputs start OFF)
Safety	`WaterController` is the only code that switches outputs; it re-checks tank, faults, caps on every pass
Logic	Never touches pins; asks `WaterController::start()` and accepts a refusal
Network	Handlers only read state or call logic methods; no handler waits on hardware or the internet

Hardware classes

All 10 classes in `Hardware.h` stay and none needs a rewrite; the additions are mostly to take values from settings instead of `Config.h` and to drive the feedback patterns.

Class (pins)	Features today	UI features it serves	Changes needed
`TimedOutput` → `Valve` ×4 (19, 18, 17, 16), `Pump` (13)	Safe OFF level before `pinMode`; `on()/off()`; own hard on-time cap; latched fault until `clearFault()`; active-high/low switch	Fault banner, Clear fault, output test, pump state	None functional. Add `onSinceMs()` for the test countdown. Keep the cap at MAX_RUN + 30 s
`SoilSensor` ×4 (36, 39, 34, 35, ADC1)	16-sample average, raw 0–4095, % from dry/wet, fault outside 100–4000, NAN on fault	Soil bar, `sf`, calibration live raw, closed loop, deficit dose	`setCal(dry, wet)` at runtime from settings; reject dry ⩽ wet. Optional: median of 3 reads to kill single spikes. Report "no rise after watering" to the mode engine (not in this class)

Class (pins)	Features today	UI features it serves	Changes needed
`TankSensor` (trig 27, echo 26 via 1k/2k)	Pulse every 500 ms, ~13 ms max wait, 2-bad → LOW, 3-good → OK, starts LOW, NAN on no echo	Tank %, tank low banner, Water now refusal, pump test refusal	`setCal(emptyCm, fullCm)`; `percent()` = (E − d)/(E − F) × 100 clamped; low threshold from settings (for example pct < 10 or d > E − 2 cm) instead of the fixed 25 cm. Optional: speed of sound from BME280 temperature (0.6 % per 10 °C)
`RainSensor` (DO 5, VCC 14)	Pull-up, wet = LOW, 2 s debounce, GPIO-powered	Raining icon, WAIT/RAIN, skip RAIN, covered	Power VCC only while sampling (for example 300 ms every 30 s) to cut electrode corrosion; it is powered permanently now. GPIO5 is a strapping pin: the pull-up keeps it HIGH at boot, keep it that way
`Environment` BME280 (I²C 21/22)	5 s period, 1× oversampling, 3 bad reads → fault, auto re-init, NAN getters	Temperature/humidity/pressure, frost and cold rules, ENV alert	Track daily Tmin/Tmax (local day) for the Hargreaves fallback. Nothing else
`Clock` DS1302 (25, 33, 32)	UTC, read once a second, valid if after 2026-01-01, `setUnix()`	Clock row, "Set clock", RTC alert, every time-based rule	Add SNTP: when Wi-Fi is up, `configTime()` and write the RTC if it is > 2 s off (every 6 h). The phone "Set clock" stays as the offline path
`Button` (4, pull-down, pressed = HIGH)	30 ms debounce, SHORT on release, LONG once while held	Button short/long actions, `btn` setting	Long press 1500 → 3000 ms. New `PRESS` event at the confirmed press-down (the spec wants LED+beep immediately). Remove the bench GPIO2 ifdef when the bench is retired
`ErrorLed` (23)	On/off only	LED feedback spec, LED output test (3 s)	Drive with LEDC PWM for the gamma-corrected sine pulse (Wi-Fi setup). Play patterns from a shared `Pattern` player

Class (pins)	Features today	UI features it serves	Changes needed
`Buzzer` (15, active, strapping pin)	Queue n × 100 ms beeps	Buzzer spec, `buzz` setting, buzzer test	Replace `beep(n)` with `play(pattern)`; patterns as on/off ms lists with a repeat flag; global mute from `buzz` (test ignores the mute)
`WaterController`	One zone at a time; valve → 1.5 s → pump; pump off → 1.5 s → valve; tank-low stop; MIN/MAX run; per-zone daily seconds; idle sweep forces outputs off; fault stop	Every watering, `pump/az/el/tot/flt`, daily limit, stop reasons	(1) daily cap from settings `cap` (clamped to MAX_ZONE_DAY_S). (2) `StopReason::DND`. (3) Report real pump seconds per run (for cost and litres). (4) A test mode (section 6, TestRunner) that allows several valves, keeps valve-before-pump, and is refused while a run is active. (5) Persist today's seconds per zone in RTC memory or NVS so a reboot cannot reset the daily cap. (6) `remainingS()` for the UI

Pin check: ADC1 for soil is right (ADC2 is dead with Wi-Fi on). Strapping pins in use: 5 (rain DO, needs HIGH at boot: pull-up OK), 15 (buzzer, chirps at boot: accepted). GPIO 2 and 12 stay unused. The handoff note's pin list is out of date; `PINS.h` is the reference.

Logic modules to add

15 modules sit between the hardware classes and the API. Each is a header with `begin()` and a non-blocking `update()`, like the hardware classes.

Module	Responsibility	Key rules
`Settings` v2	All global and per-zone values, NVS, clamp on load and save	`VERSION` 2 (resets once). Drops `winter`, `runS`, `startMin/endMin`, `coveredUntil` date → `cov` bool. Adds name, cap, rainmm, qto, lat/lon, flow, tankE/F, soilCal ×4, cost fields, btn, buzz, per-zone name/plant/soil/spr/cold/cov. Raising `frost` raises zone `cold` values below it
`CalendarStore`	Watering entries `{sid,d,s,sec,dt}` per zone; DND windows `{sid,d,s,e,dt,de}` per zone and global	Fixed capacity (proposal: 16 watering + 8 DND per zone, 8 global), `sid` unique and never reused within a boot; LittleFS JSON or NVS blob per zone; prune one-offs older than 7 days

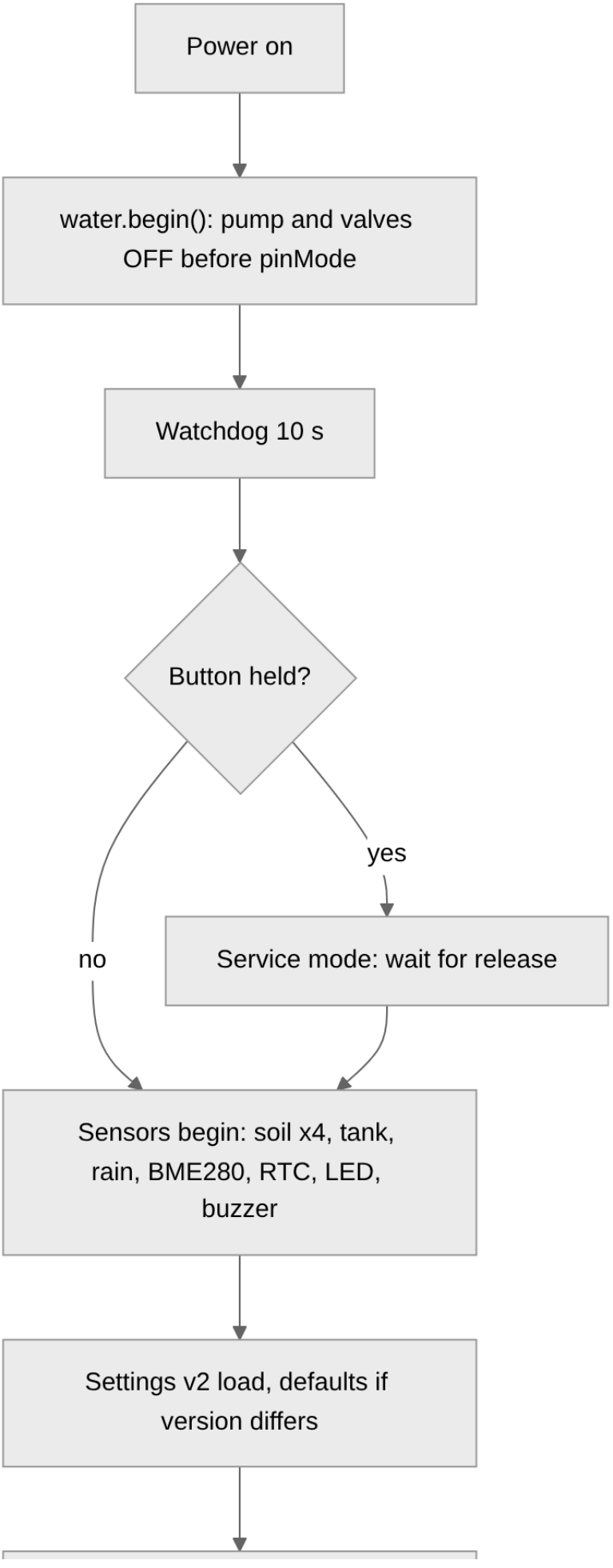
Module	Responsibility	Key rules
`DndManager`	One-off DND until a time (zone, global, `FOREVER`) and the weekly/one-off windows	`active(zone, t)`; windows past midnight when `e ≤ s`; setting DND that covers the running zone stops it with result DND; expired one-offs clear themselves
`Rules`	One place for precedence and zone status	`globalBlock()` = ALL_OFF › FROST › TANK_LOW (frost skipped when the BME280 is faulted); `zoneState()` → ON/OFF/DND/WATERING/QUEUED/ALLOFF/FAULT/WAIT + why (flowchart 7.6)
`CalendarScheduler`	Fires due entries, decides run / skip, predicts `next`	Each entry fires once per occurrence (persist last-fired per sid/day so a reboot cannot repeat it); blocked → wait in queue up to `qto`, then SKIP_QUEUE_TIMEOUT; DND at start → SKIP_DND; `next` scans 8 days
`ModeEngine`	Turns an entry into a dose for the zone's mode	Schedule: `sec`. Closed loop: water only if soil < lo, dose = deficit to hi. Open loop: same with the water-balance estimate. Sensor fault or 3 runs without a rise → fall back to Schedule length (15 min if none) and raise the `sf` warning
`Deficit`	Dose formula shared by closed loop, open loop, Water now "Fill the deficit" and the button	available mm = (fc − wp)/100 × root × 10; deficit = (hi − soil)/100 × available; sec = deficit / (eff/100) / rate × 3600, rounded to 10 s, capped at 3600 s and at the daily remainder (button: 900 s)
`WaterBalance`	Open-loop soil estimate per zone	Daily: soil −= ET_0 × Kc / available × 100; += effective rain (if not covered) and += applied mm after each run; clamp 0–100 (see section 9 on what the % means). Persist once a day and after each run
`WeatherClient`	Open-Meteo daily tmax, tmin, et0, rain, rain chance for today and tomorrow	Fetch every 1–3 h in its own FreeRTOS task (HTTPS blocks for seconds); cache `t`, `ok`; on failure keep the last good data; Hargreaves ET_0 from BME280 Tmin/Tmax + latitude when there is no forecast. Forecast rain ≥ `rainmm` → skip with reason RAIN
`RunQueue`	Holds waiting runs (scheduled and Water now)	FIFO, one entry per zone; next starts only when the previous one ended DONE; any stop by user, tank, fault or DND clears it (matches the mock); entries older than `qto` are dropped and logged
`CostMeter`	Pump seconds per period d/w/m/season/y	Add each run's pump seconds; roll over at local midnight, Monday 00:00, 1st of month, 1 Jan; season starts when All off is switched off; write COST rows for periods in `clog`; `cr=1` and `/api/cost/reset` zero the counters. Persist to NVS on every run end (not every second)
`AlertMonitor`	Edge-detects TANK, ENV, SOIL n, FAULT, RTC, WIFI (rssi < −80 dBm)	Write ALERT rows on change (E/W, I when cleared); debounce Wi-Fi and soil for about 60 s so a flicker does not fill the log
`RunLog`	v2 CSV `utc,local,zone,mode,soil0,soil1,sec,litres,result` + ALERT + COST rows	Rotation at 256 KB, `tail(n)`, per-zone last 5 rows and `last` run for the state (cache them in RAM, do not scan the file per poll)
`TestRunner`	Output tests	Valve 60 s each (several at once), pump follows the open valves and ends 2 s before the last closes, pump alone 10 s, LED 3 s, buzzer pattern; refused while a run is active; stopping the last valve stops the pump; uses `WaterController` test mode

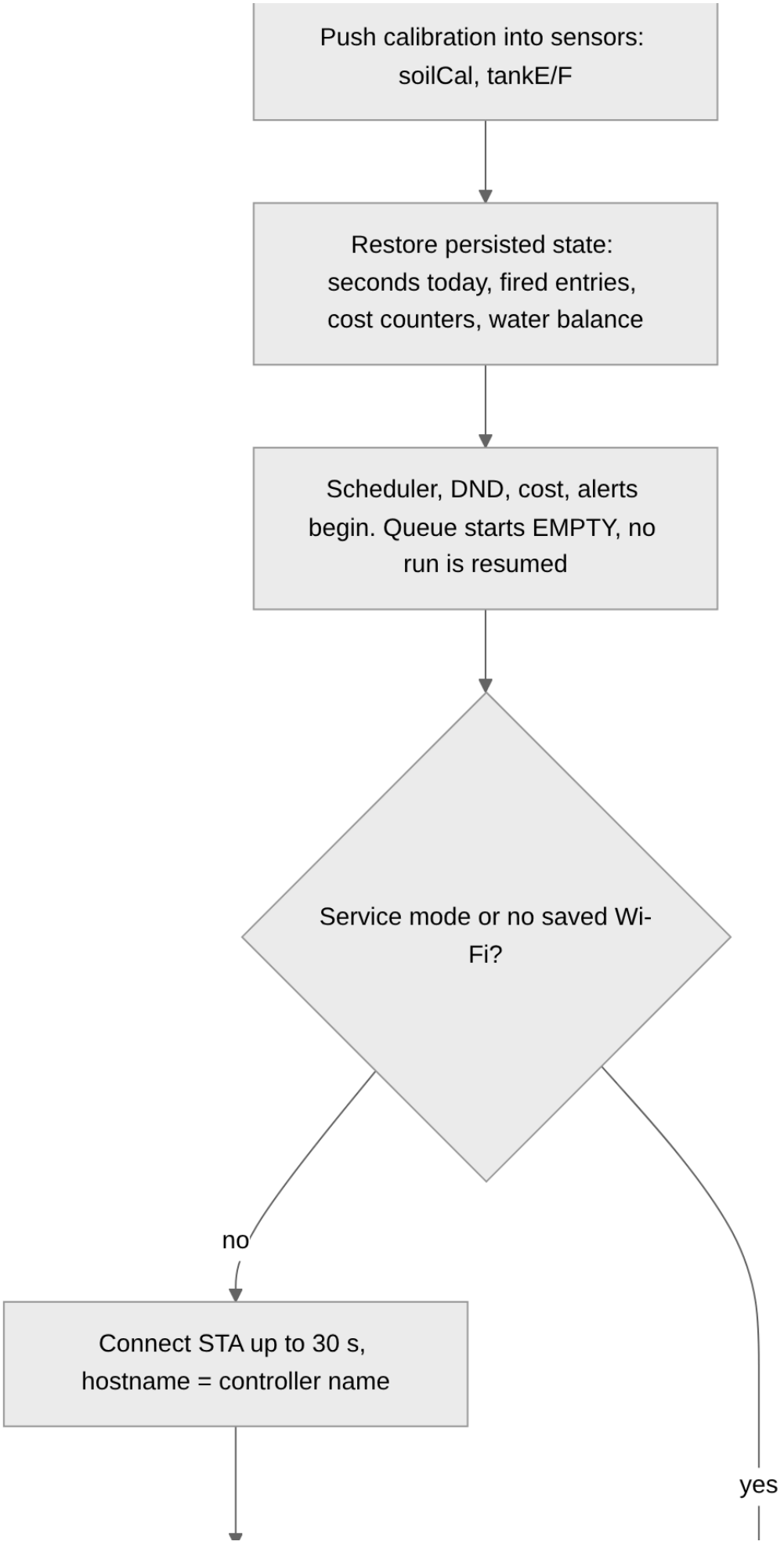
Module	Responsibility	Key rules
`ButtonActions` + `Feedback`	Button short/long behaviour, LED and buzzer patterns	Exactly the "Button & feedback" spec in API-v2.md (flowchart 7.8); `btn` = 0 disables the short press only; `buzz` = 0 mutes everything

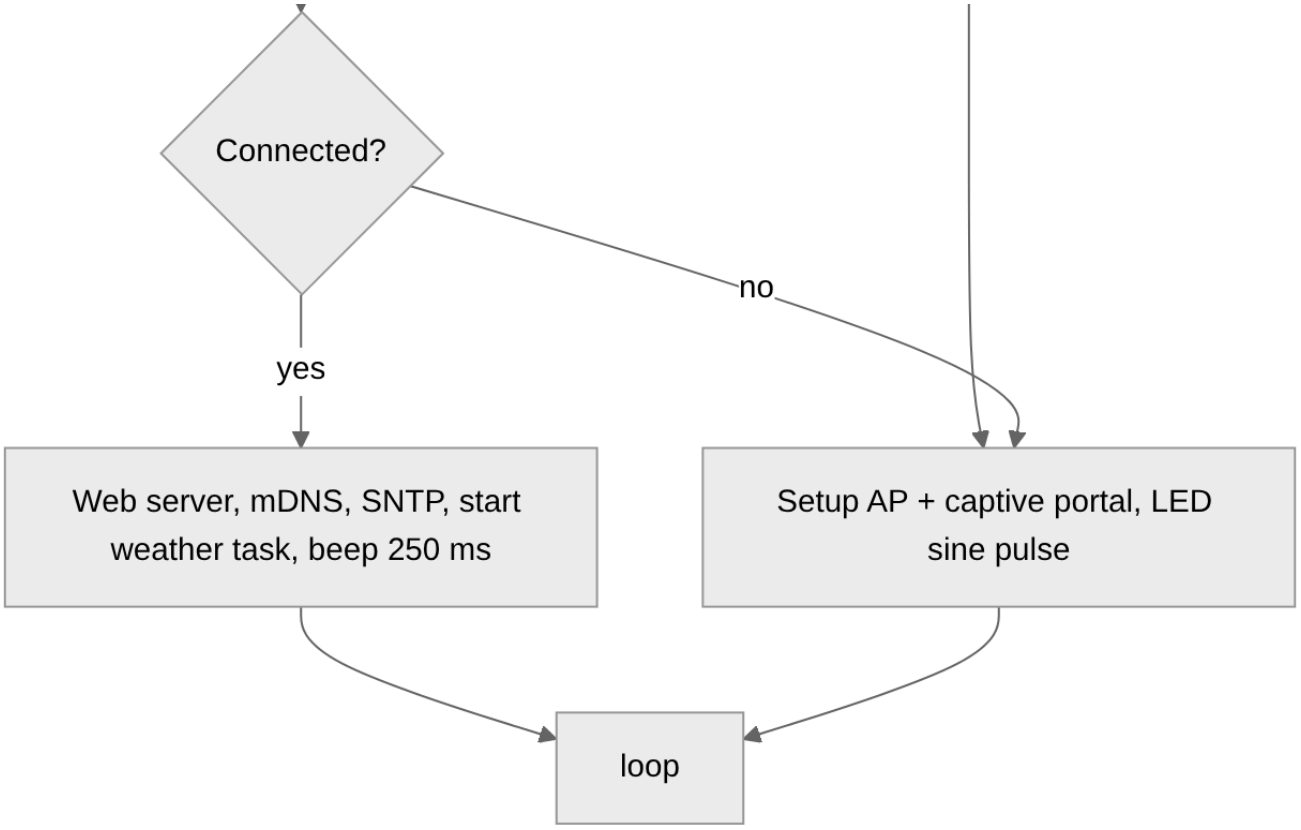
Flowcharts: how it should work

Twelve flows cover every behaviour the UI expects. Codes in `r:` are the exact strings the UI translates.

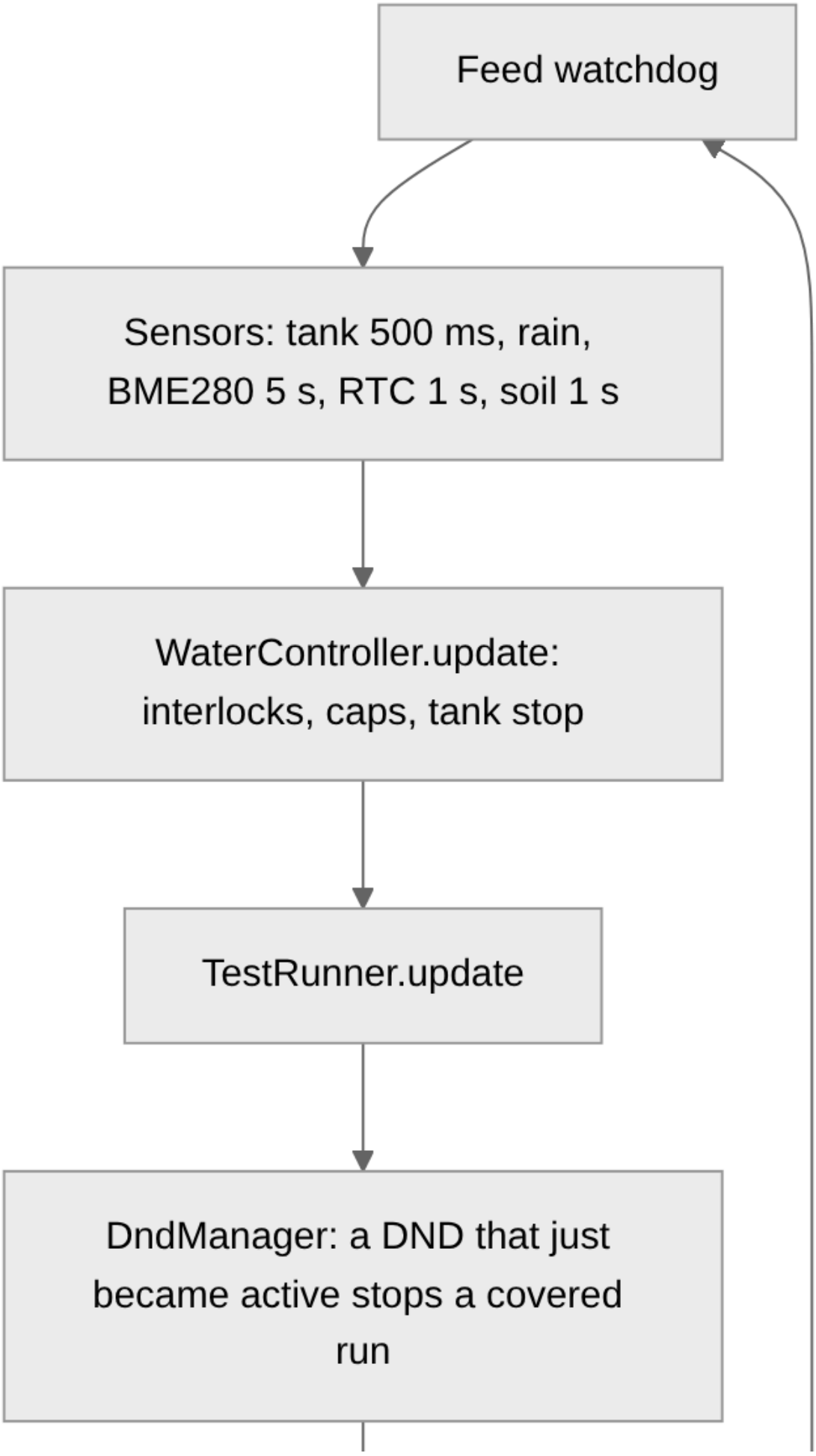
7.1 Boot

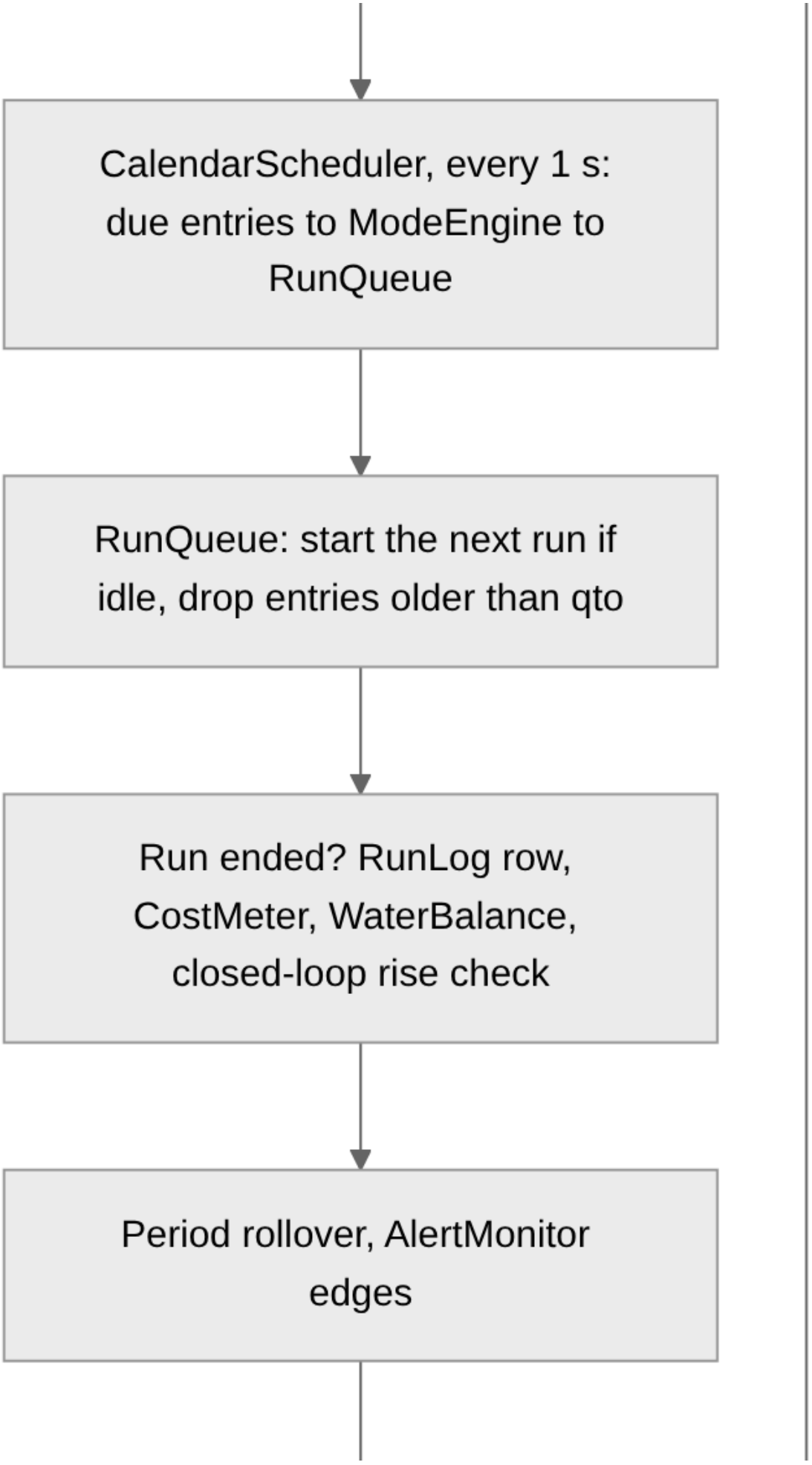

Power on
water.begin(): pump and valves OFF before pinMode
Watchdog 10 s
Button held?
yes
no
Service mode: wait for release
Sensors begin: soil x4, tank, rain, BME280, RTC, LED, buzzer
Settings v2 load, defaults if version differs

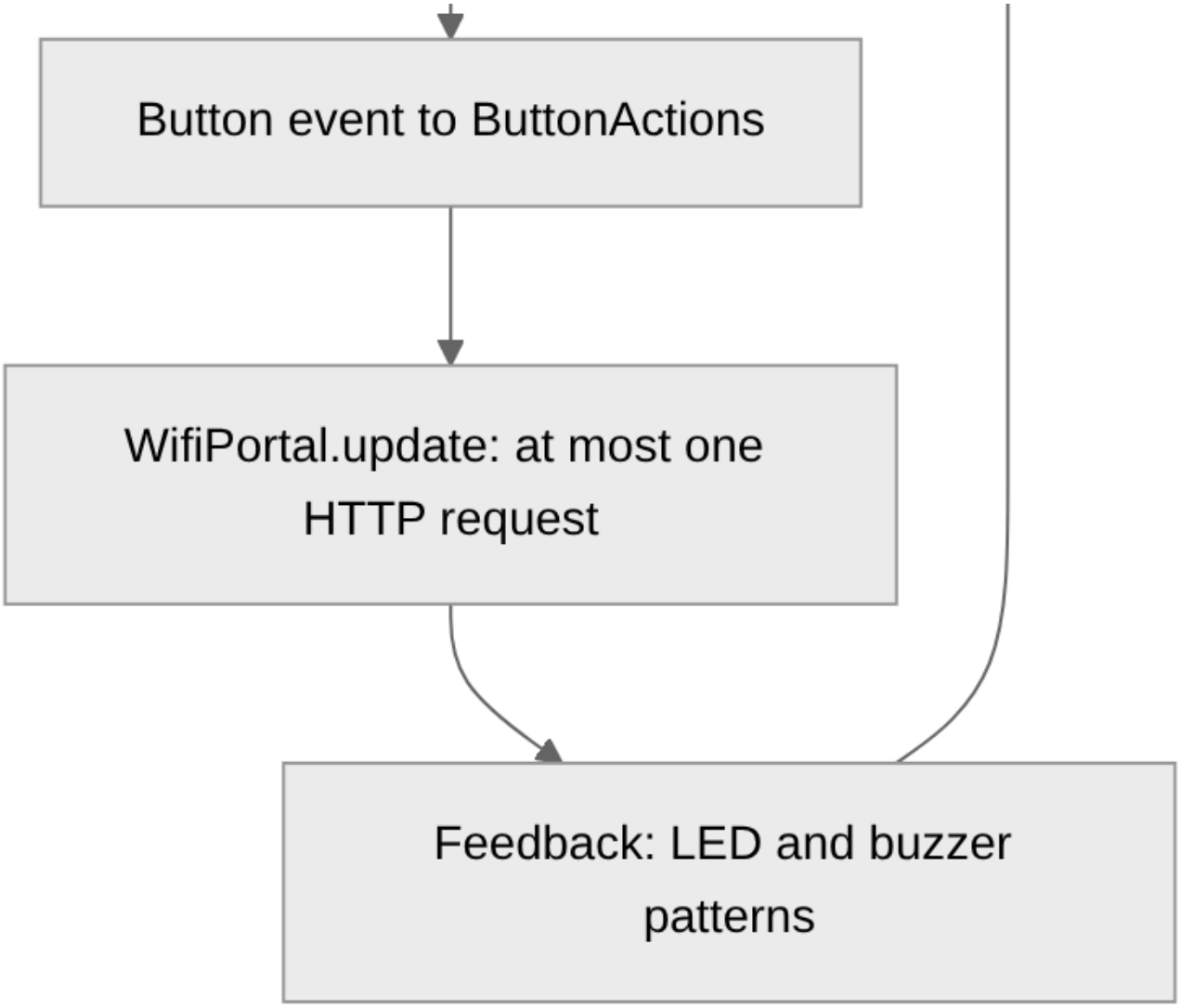

Push calibration into sensors: soilCal, tankE/F
Restore persisted state: seconds today, fired entries, cost counters, water balance
Scheduler, DND, cost, alerts begin. Queue starts EMPTY, no run is resumed
Service mode or no saved Wi-Fi?
no
yes
Connect STA up to 30 s, hostname = controller name



7.2 One pass of the main loop


Feed watchdog
Sensors: tank 500 ms, rain, BME280 5 s, RTC 1 s, soil 1 s
WaterController.update: interlocks, caps, tank stop
TestRunner.update
DndManager: a DND that just became active stops a covered run

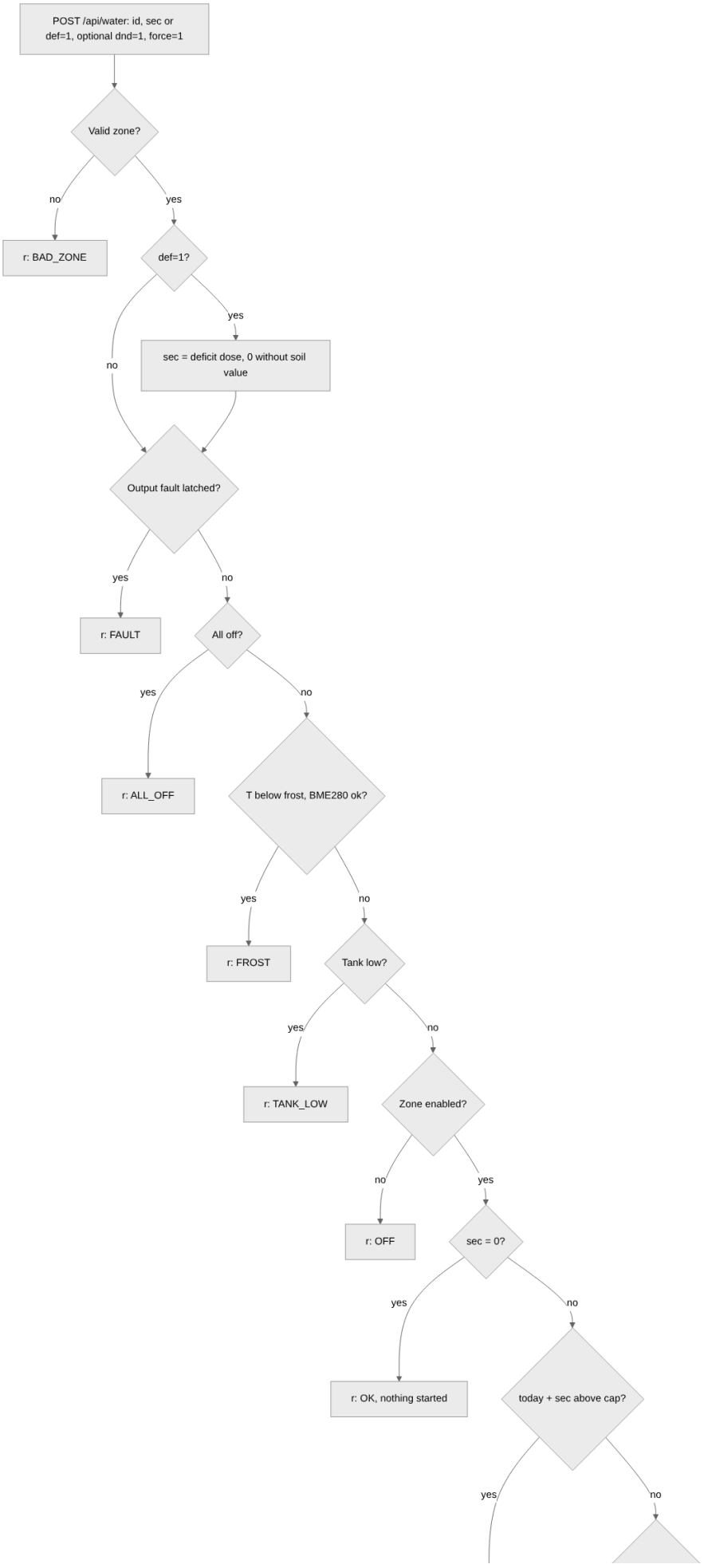

CalendarScheduler, every 1 s: due entries to ModeEngine to RunQueue
RunQueue: start the next run if idle, drop entries older than qto
Run ended? RunLog row, CostMeter, WaterBalance, closed-loop rise check
Period rollover, AlertMonitor edges

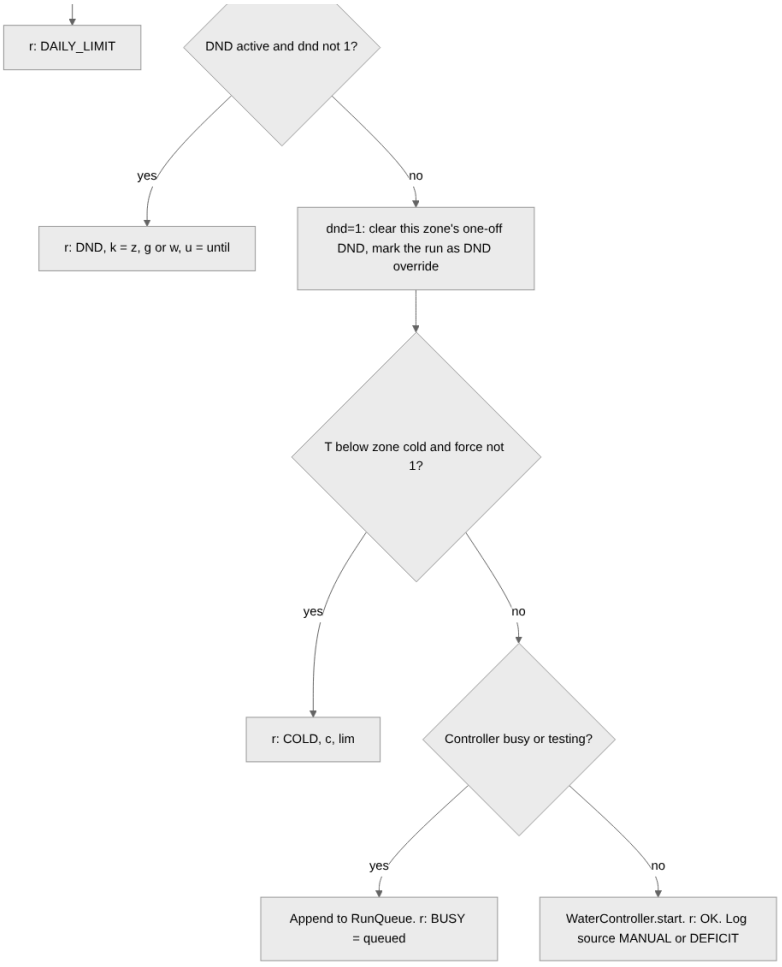


Budget: one pass under 20 ms. The HC-SR04 wait (up to 13 ms) and the soil sampling (4 × 3.2 ms) are the only blocking reads; the log download and HTTPS weather fetch must not run inside the loop pass.

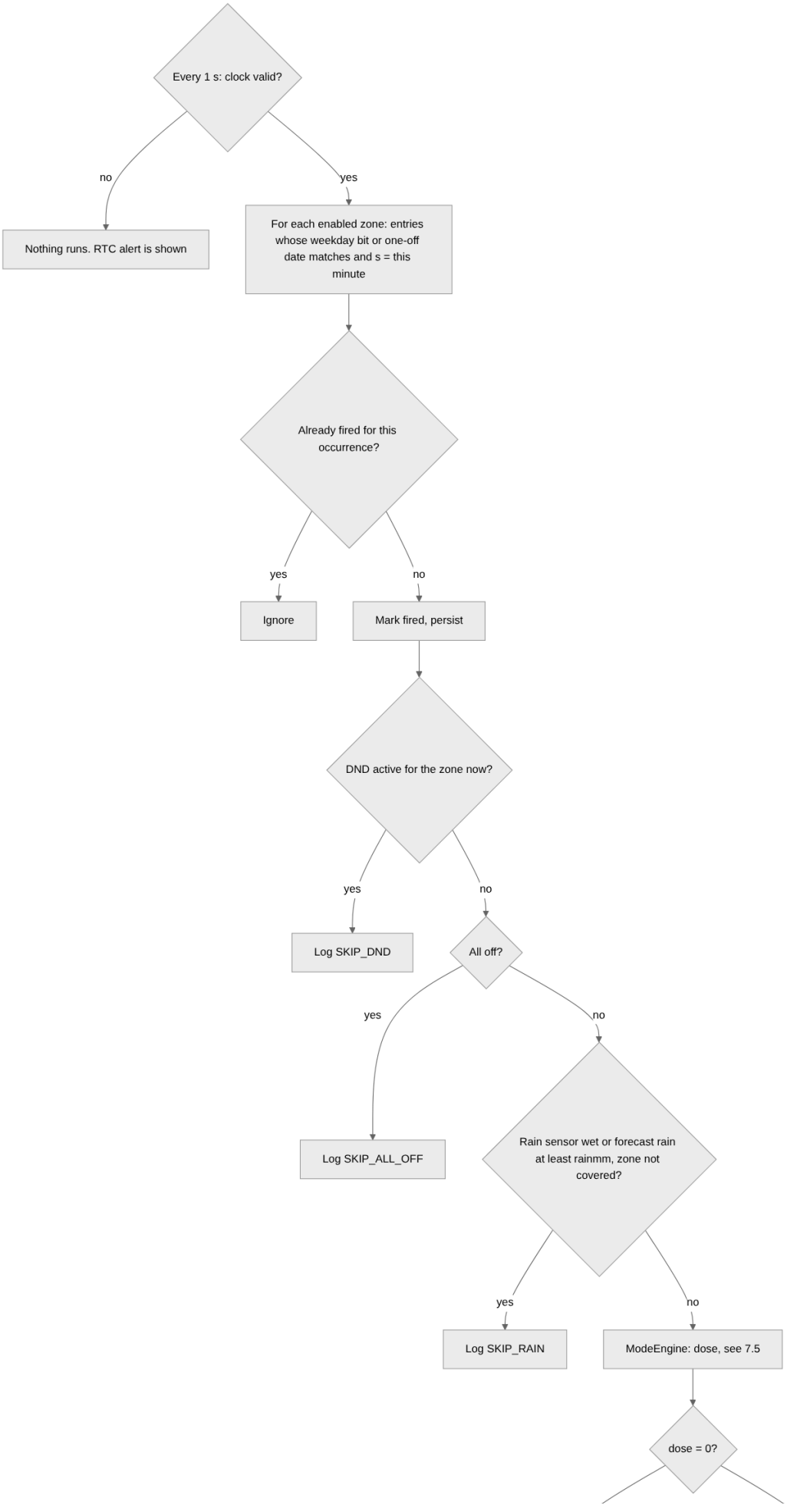
7.3 Water now (`POST /api/water`)

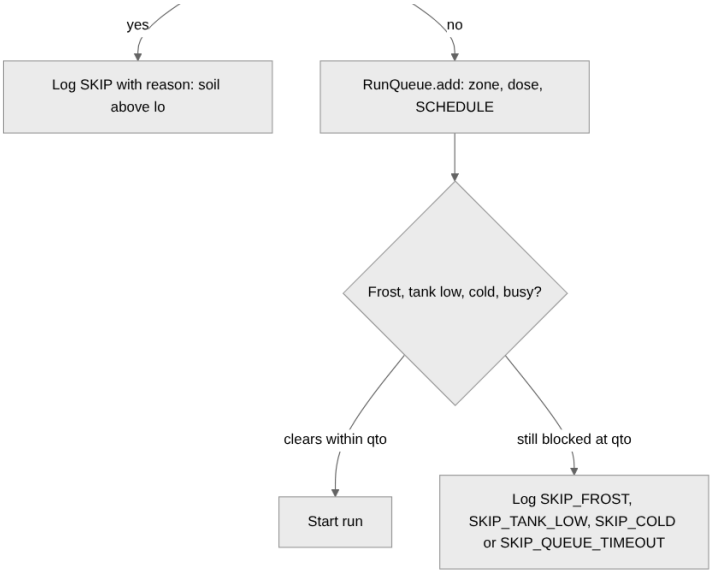
The order follows the mock and the handoff precedence: hard global blocks first, then the zone, then DND and cold, which the user can override.


POST /api/water: id, sec or def=1, optional dnd=1, force=1
Valid zone?
no
yes
r: BAD_ZONE
def=1?
yes
no
sec = deficit dose, 0 without soil value
Output fault latched?
yes
no
r: FAULT
All off?
yes
no
r: ALL_OFF
T below frost, BME280 ok?
yes
no
r: FROST
Tank low?
yes
no
r: TANK_LOW
Zone enabled?
no
yes
r: OFF
sec = 0?
yes
no
r: OK, nothing started
today + sec above cap?
yes
no



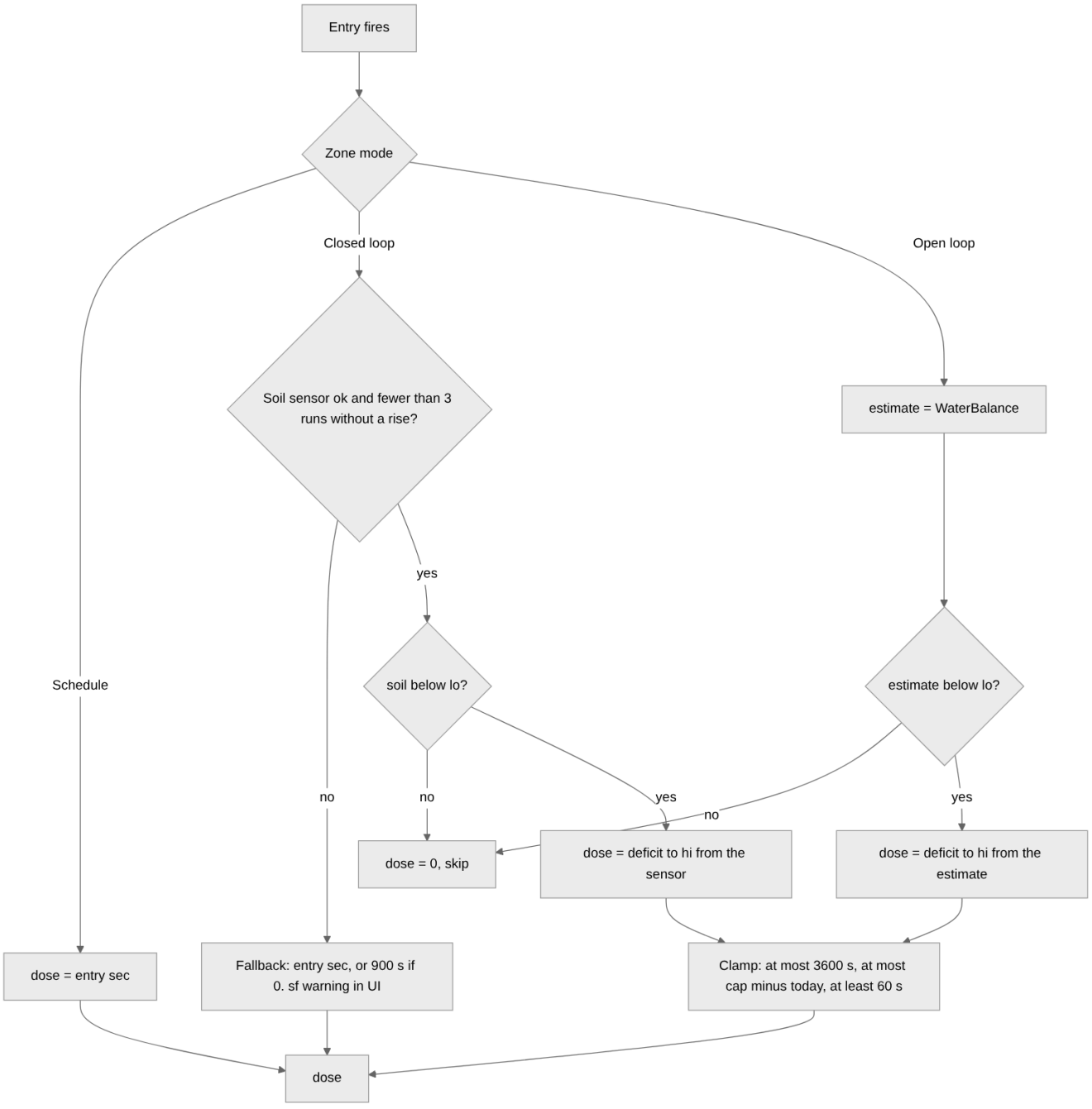
7.4 A calendar entry becomes due


Every 1 s: clock valid?
no
yes
Nothing runs. RTC alert is shown
For each enabled zone: entries whose weekday bit or one-off date matches and s = this minute
Already fired for this occurrence?
yes
no
Ignore
Mark fired, persist
DND active for the zone now?
yes
no
Log SKIP_DND
All off?
yes
no
Log SKIP_ALL_OFF
Rain sensor wet or forecast rain at least rainmm, zone not covered?
yes
no
Log SKIP_RAIN
ModeEngine: dose, see 7.5
dose = 0?



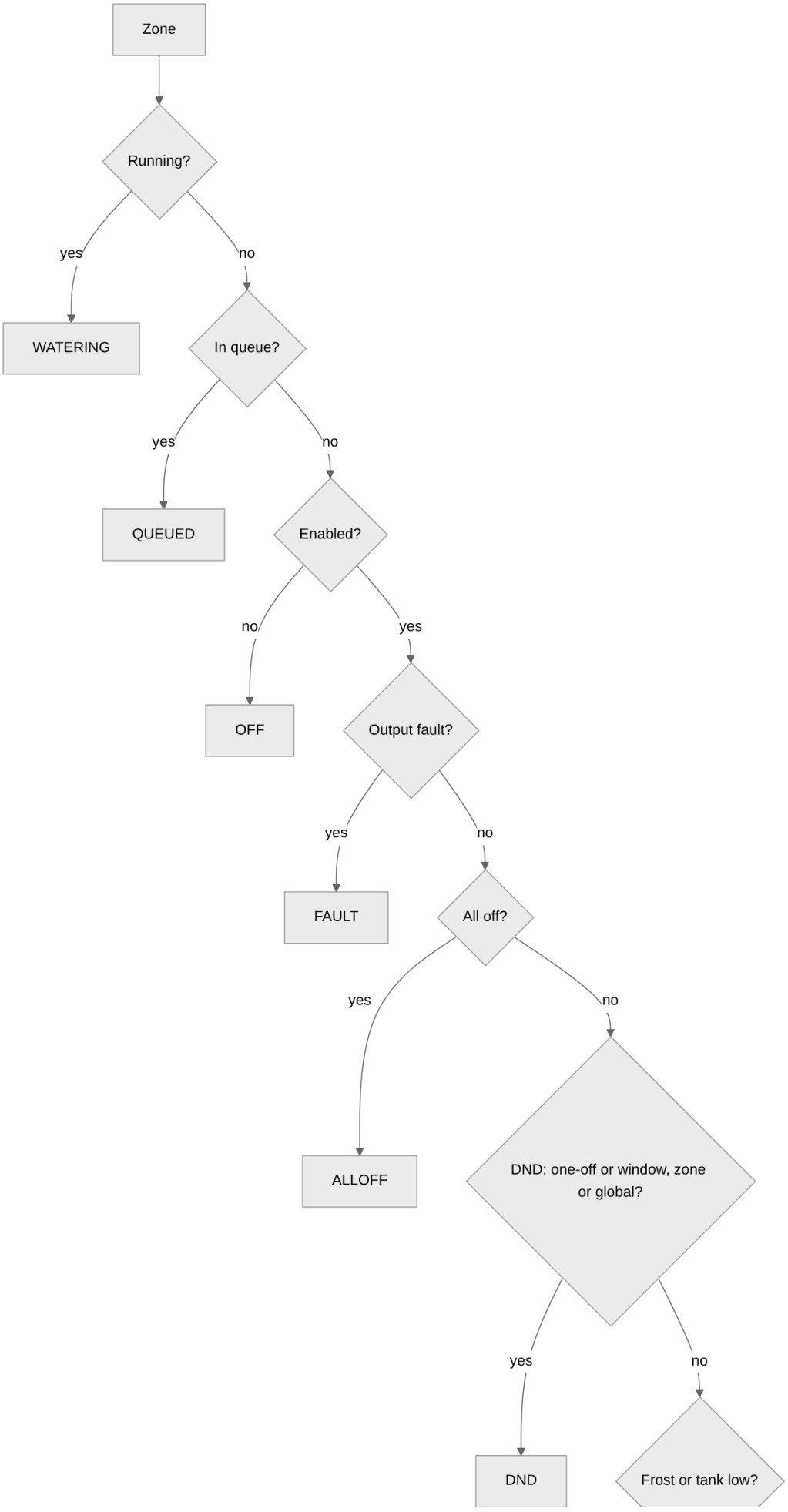
After a power cut: an occurrence missed by less than `qto` is queued late on boot; an older one is logged `SKIP_WINDOW_MISSED` (the UI already has that word).

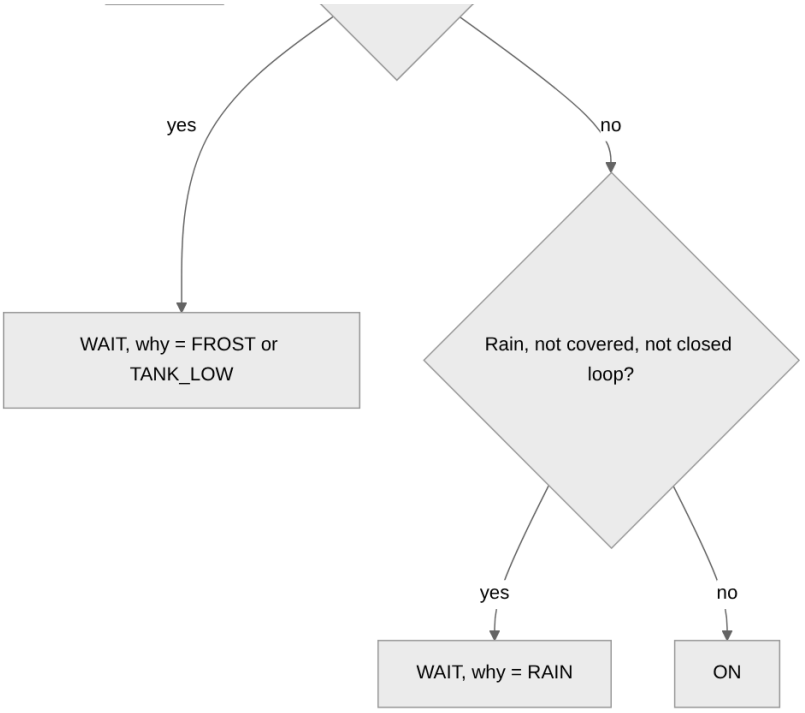
7.5 Dose per mode



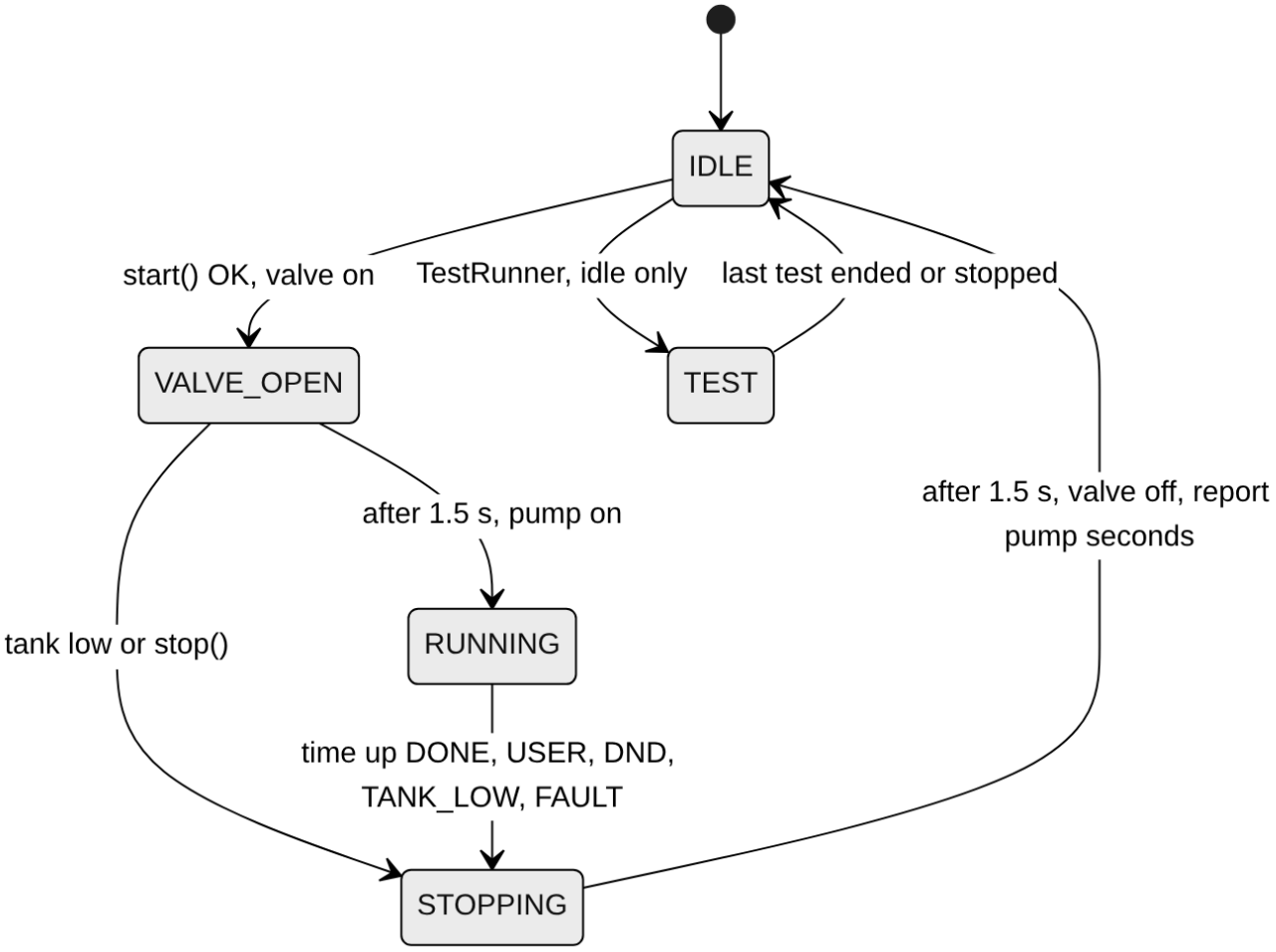
Rise check after each closed-loop run: if soil1 − soil0 is under 2 points, count a miss; 3 misses in a row switch the zone to the fallback until the sensor shows a rise again.

7.6 Zone status `st` and `why` (first match wins)


Zone
Running?
yes
no
WATERING
In queue?
yes
no
QUEUED
Enabled?
no
yes
OFF
Output fault?
yes
no
FAULT
All off?
yes
no
ALLOFF
DND: one-off or window, zone or global?
yes
no
DND
Frost or tank low?

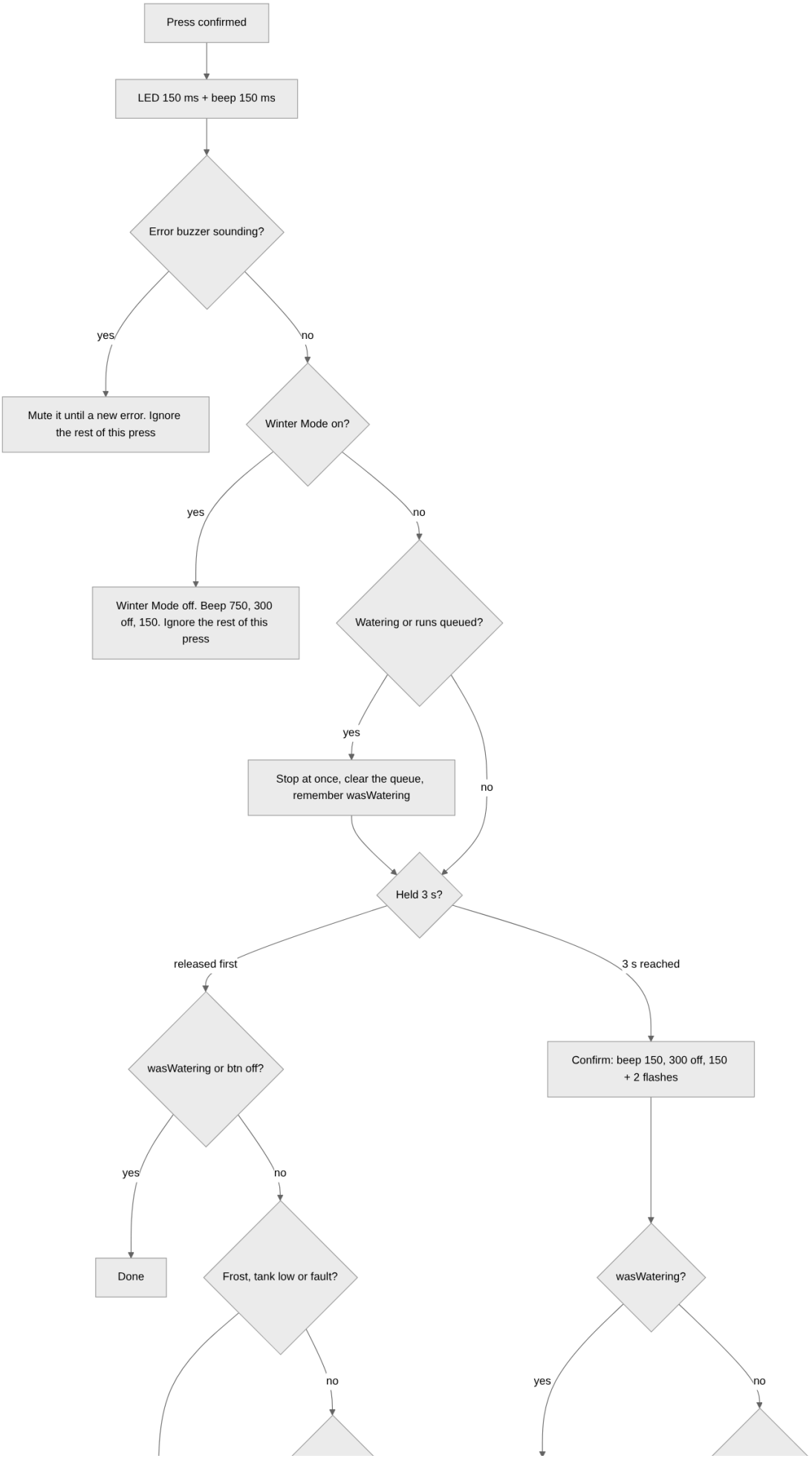


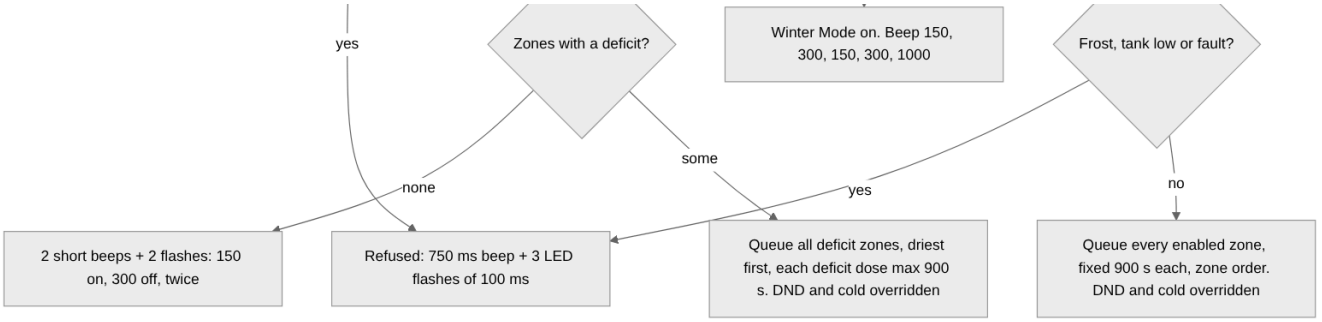
7.7 WaterController states



In IDLE every output is forced off each pass. In TEST only the outputs the TestRunner owns may be on, with the same valve-before-pump and pump-before-valve delays, the tank-low stop and the hard caps.

7.8 Button (API-v2.md spec)

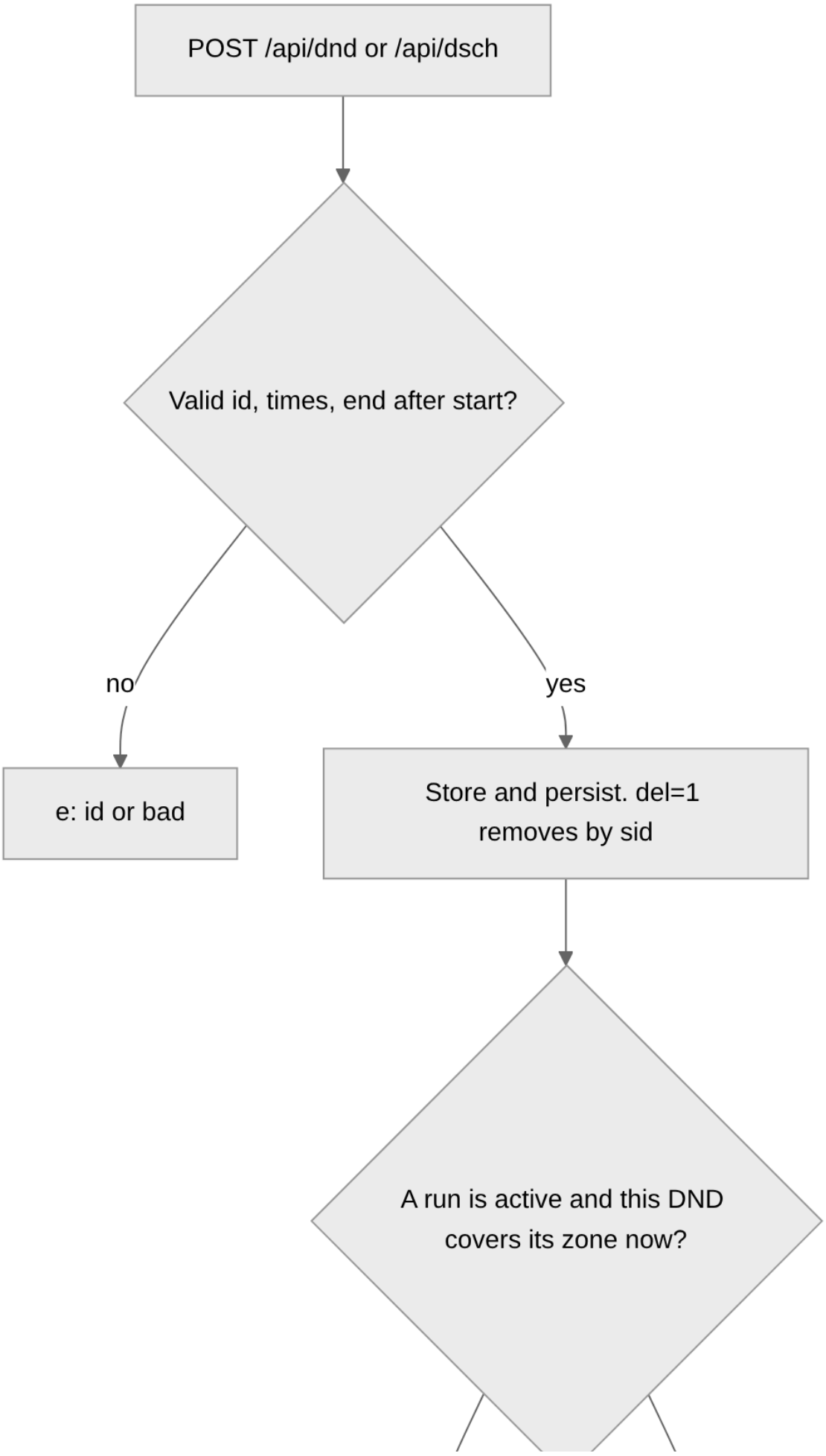

Press confirmed
LED 150 ms + beep 150 ms
Error buzzer sounding?
yes
no
Mute it until a new error. Ignore the rest of this press
Winter Mode on?
yes
no
Winter Mode off. Beep 750, 300 off, 150. Ignore the rest of this press
Watering or runs queued?
yes
no
Stop at once, clear the queue, remember wasWatering
Held 3 s?
released first
3 s reached
wasWatering or btn off?
Confirm: beep 150, 300 off, 150 + 2 flashes
yes
no
Done
Frost, tank low or fault?
wasWatering?
no
yes
no

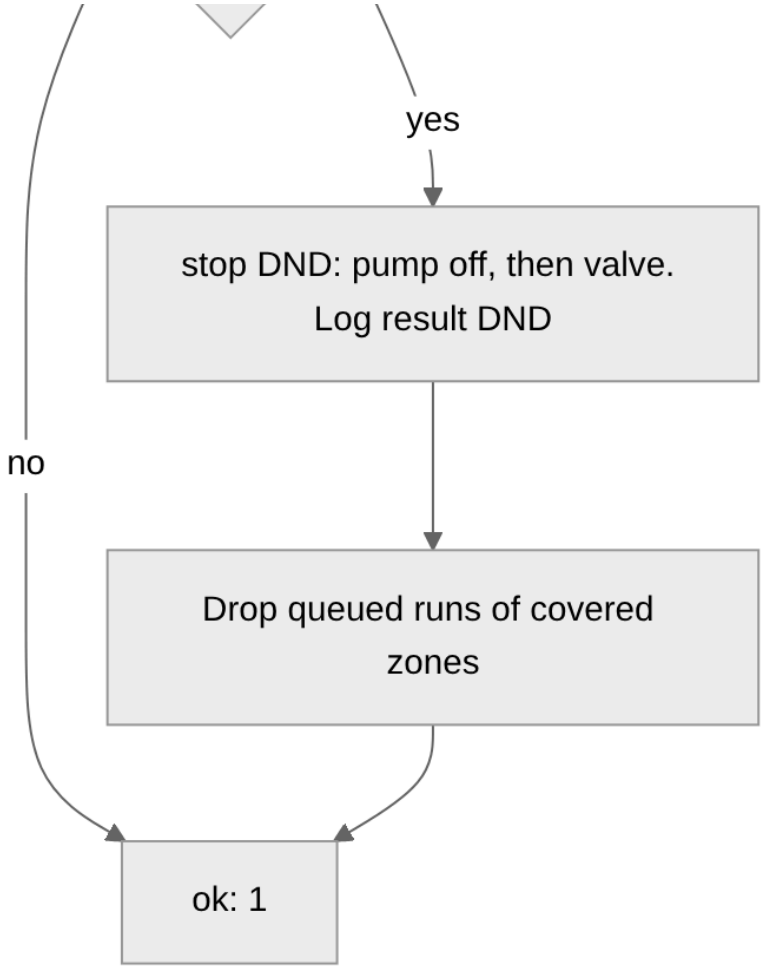


Decided 30 Oct:

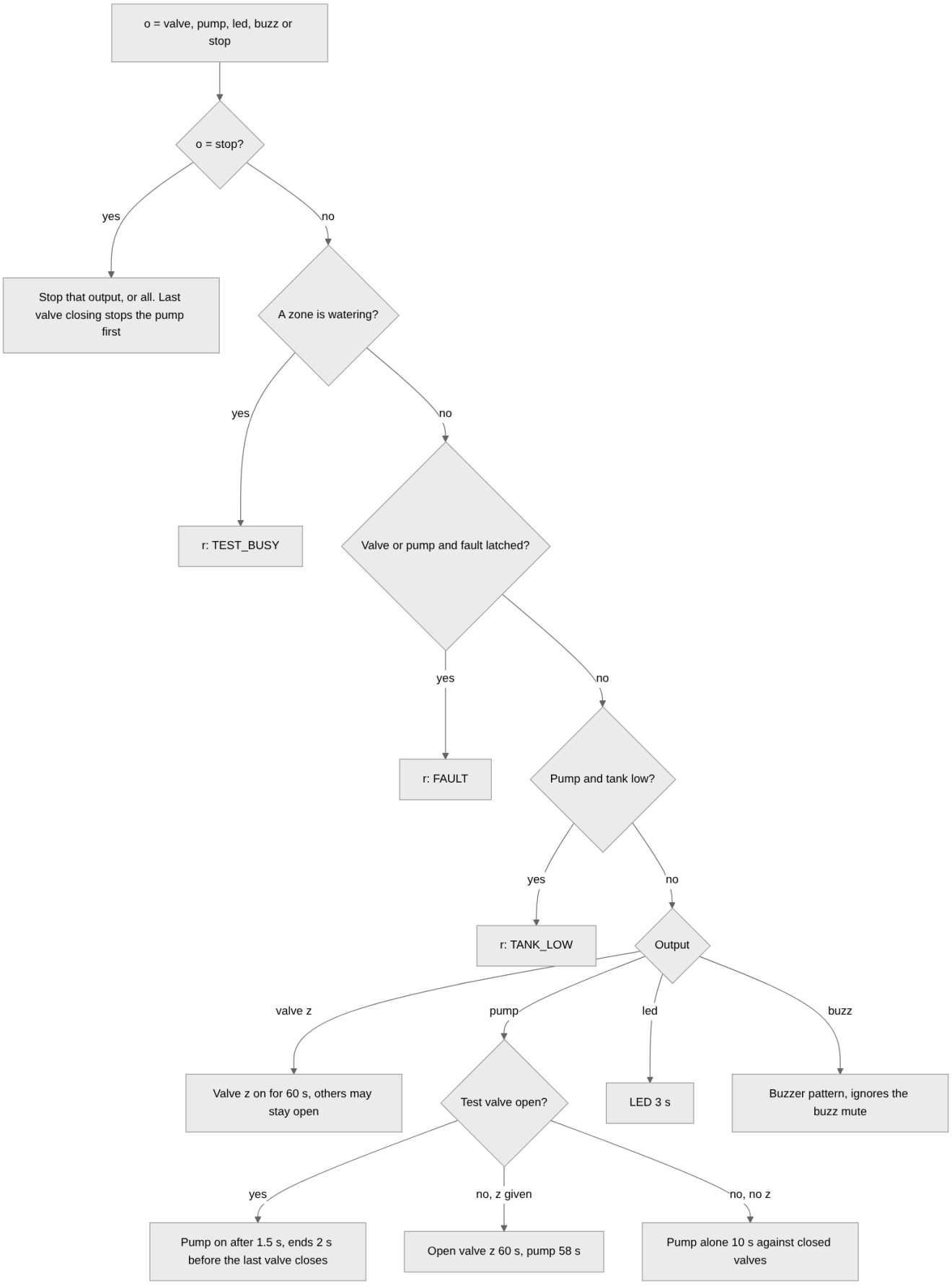
- **Short press, idle:** every zone with a deficit, driest (largest deficit in mm) first, each dose capped at 15 min; the UI shows the capped length. Zones with no known soil value (sensor fault, Schedule mode without a sensor) are skipped; Open loop uses its estimate.
- **Long press, idle:** every enabled zone, a fixed 15 min, ignoring sensors: a manual full cycle.
- **While watering:** the stop happens at press-down; still held at 3 s → Winter Mode on as well.
- **Overrides:** the button skips DND and the cold limit. Frost, tank low and fault refuse; the daily limit trims or skips that zone. Log source MANUAL (long press) or DEFICIT (short press).
- **Serious error** = safety cut-off, tank low or clock lost. LED 300 on / 200 off until it clears; buzzer 150/300 ×3 then 3000 off, repeating, until a button press mutes it. A mute press does nothing else; a new error beeps again.
- **Cycle beeps:** 250 ms when the first zone of a cycle starts and when the last one ends; none between zones. Wi-Fi connected 250 ms.
- **LED when no pattern plays:** serious error › Wi-Fi setup (sine pulse, 1200 ms) › Winter Mode (1500 ms on every 5 s) › off. `buzz = 0` mutes every sound except the output test.

7.9 Setting DND


POST /api/dnd or /api/dsch
Valid id, times, end after start?
no
yes
e: id or bad
Store and persist. del=1 removes by sid
A run is active and this DND covers its zone now?

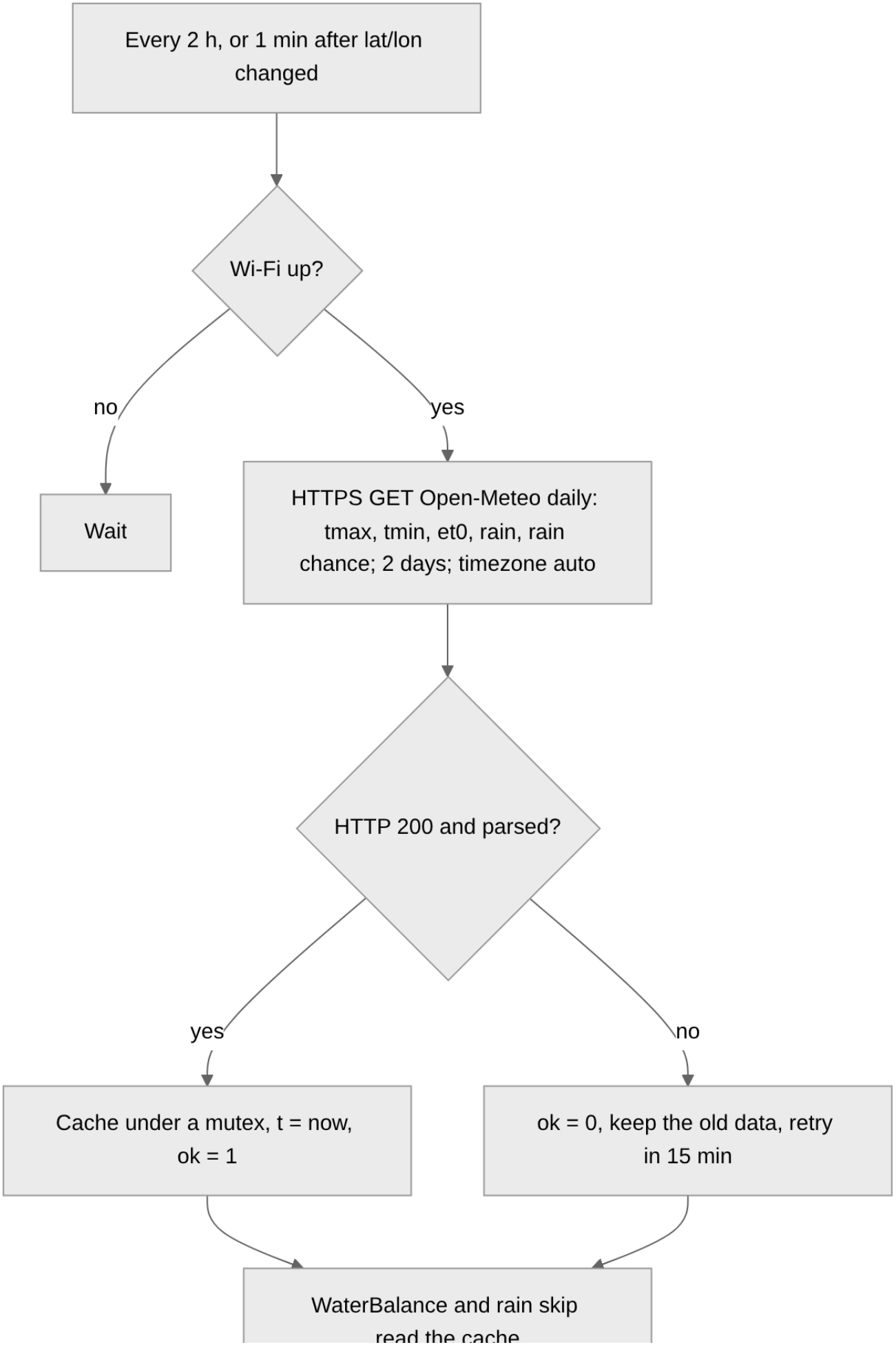


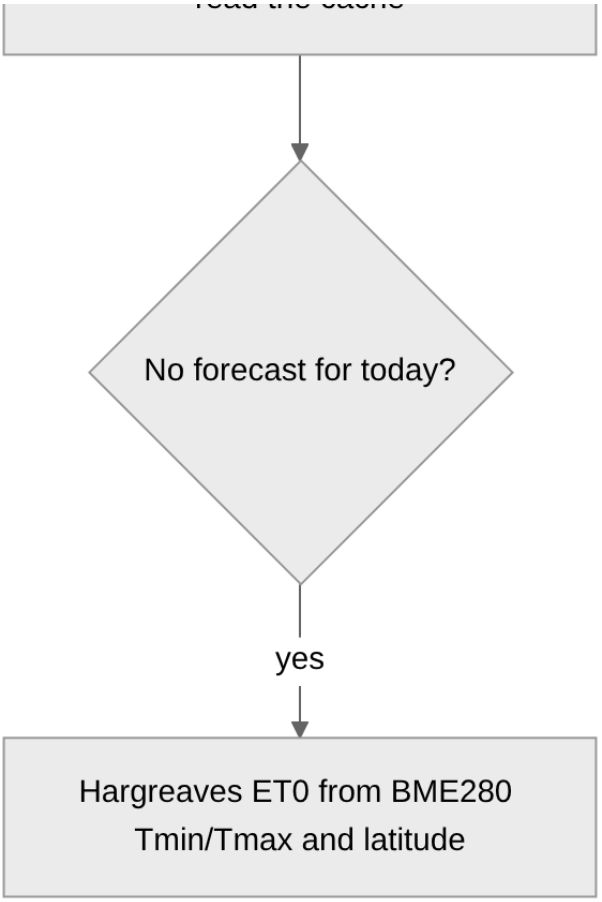
7.10 Output test (`POST /api/test`)



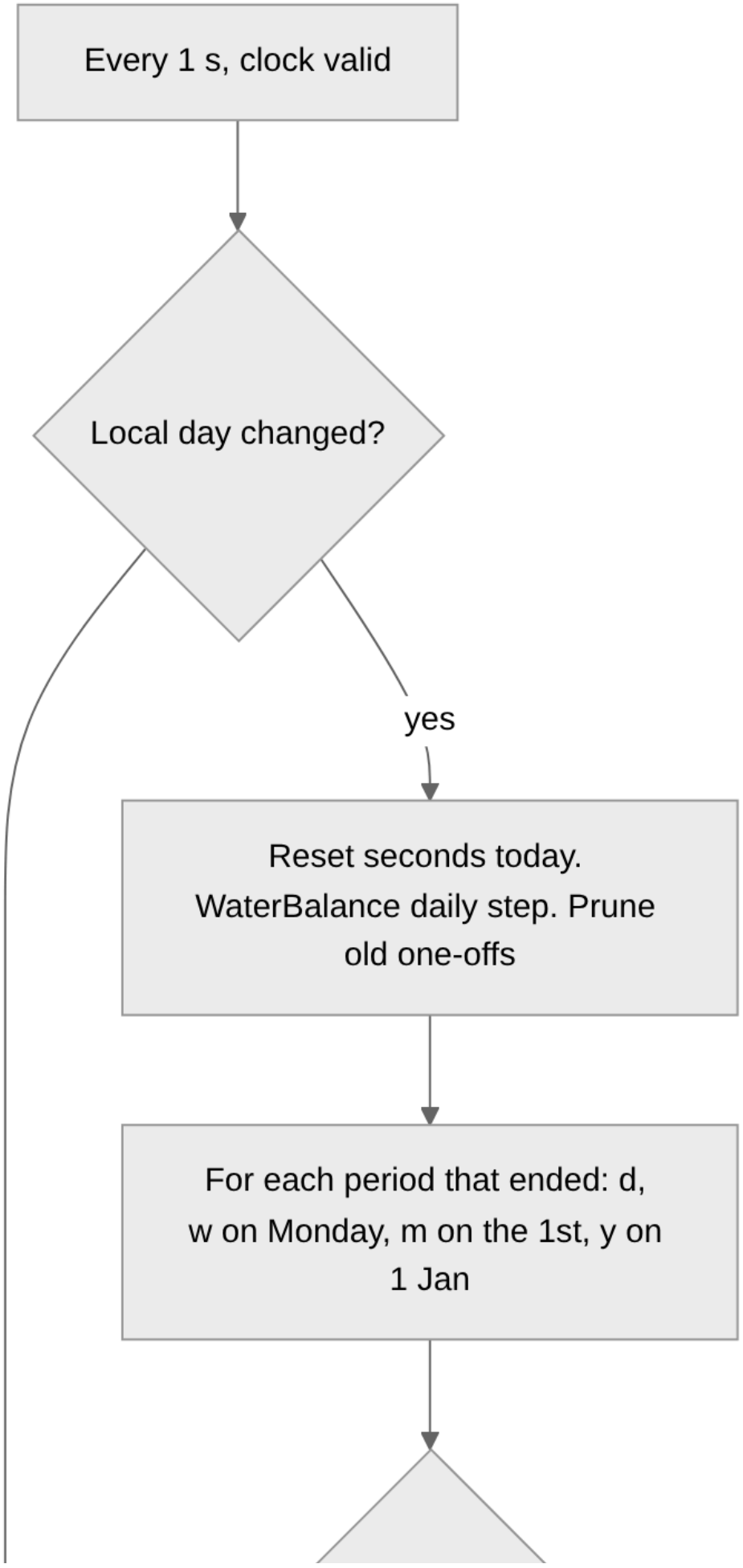
While a test runs, the RunQueue treats the controller as busy: scheduled runs and Water now wait, they are not refused.

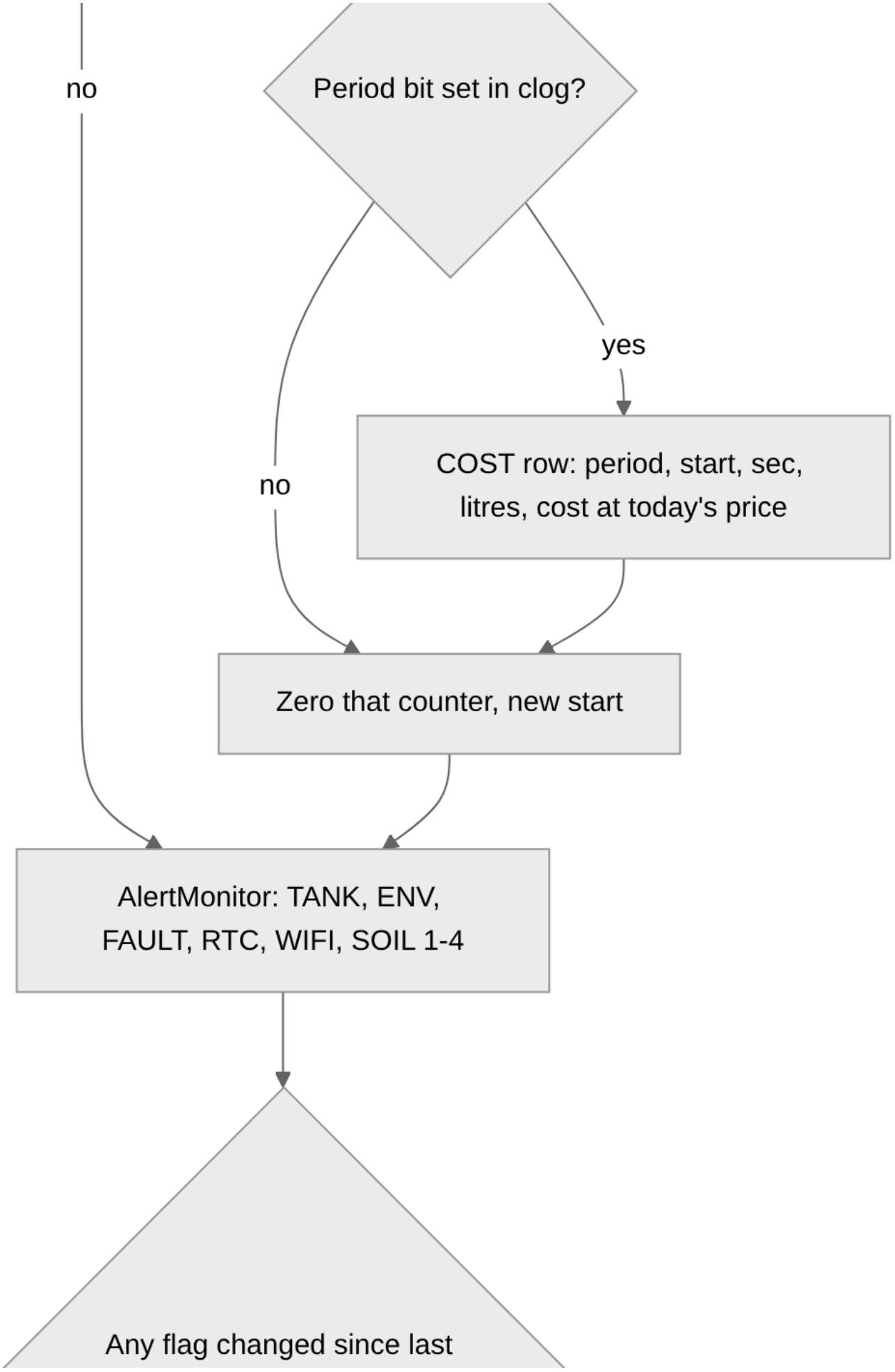
7.11 Weather fetch (own FreeRTOS task)

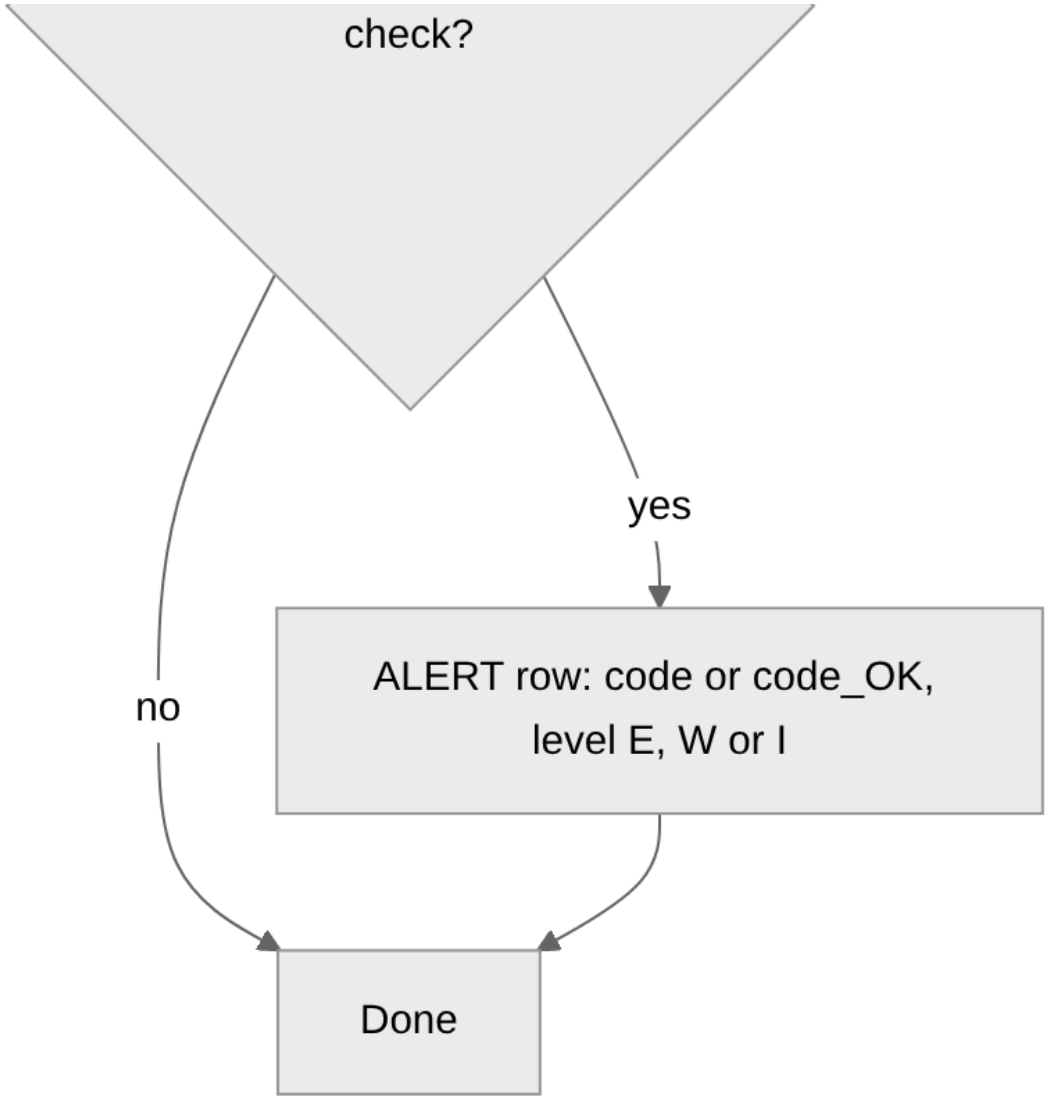

Every 2 h, or 1 min after lat/lon changed
Wi-Fi up?
no
yes
Wait
HTTPS GET Open-Meteo daily: tmax, tmin, et0, rain, rain chance; 2 days; timezone auto
HTTP 200 and parsed?
yes
no
Cache under a mutex, t = now, ok = 1
ok = 0, keep the old data, retry in 15 min
WaterBalance and rain skip
read the cache



7.12 Periodic jobs: day rollover, cost, alerts


Every 1 s, clock valid
Local day changed?
yes
Reset seconds today. WaterBalance daily step. Prune old one-offs
For each period that ended: d, w on Monday, m on the 1st, y on 1 Jan


no
Period bit set in clog?
yes
COST row: period, start, sec, litres, cost at today's price
no
Zero that counter, new start
AlertMonitor: TANK, ENV, FAULT, RTC, WIFI, SOIL 1-4
Any flag changed since last



Data storage

Everything except the log fits in NVS (about 3 KB of the default 20 KB partition); keep one blob per key so one write never rewrites everything, and write only on change.

NVS key / file	Contents	Approx. size	Written when
`cfg/g`	Version, name (hostname), off, frost, cap, rainmm, qto, lat, lon, flow, tankE/F, cost mode and prices, currency, clog, btn, buzz	120 B	A setting changes
`cfg/z1…z4`	Name (96 B UTF-8), en, m, cold, cov, plant, soil, spr	4 × 150 B	A zone setting changes

NVS key / file	Contents	Approx. size	Written when
`cfg/cal`	Soil dry/wet ×4	16 B	Calibration changes
`cal/w1…w4`	Watering entries, max 16 per zone, 12 B each	4 × 200 B	An entry is saved or deleted
`cal/d0…d4`	DND windows, global + per zone, max 8 each, 16 B each	5 × 130 B	A window is saved or deleted
`rt/dnd`	One-off DND until, global + 4 zones	20 B	DND set or cleared
`rt/run`	Seconds today per zone + local day, last-fired day per entry sid	150 B	Run end, entry fired
`rt/cost`	5 counters `{sec, f}`	40 B	Run end, period rollover, reset
`rt/wb`	Open-loop estimate per zone + date	24 B	Daily step, run end
`wifi/ssid, pass`	Credentials (WifiPortal)	100 B	Setup portal
LittleFS `/log.csv`, `/log_old.csv`	v2 log, rotate at 256 KB	⩽ 512 KB	Run end, skip, alert, cost row
RAM only	Queue, live sensor values, weather cache, running tests, alert previous flags, last-run and last-5 per zone cache	~4 KB	—

NVS write budget: a few writes per run and per setting change, roughly 20–50 a day, far below the flash wear limit. A version bump (1 → 2) drops the old blob once; the experiment should start after that.

Inconsistencies and decisions needed

Items 1–5 change the experiment data or the safety behaviour, so decide them before coding starts; the rest can follow.

#	Issue found	Proposal
1	**Soil % has two meanings.** The sensor gives 0 % = dry air, 100 % = in water. The deficit formula and the mock treat it as % of plant-available water (0 = wilting point, 100 = field capacity). Preset `lo/hi` (35/50) only make sense in one of them	Pick one and write it in the paper. Simplest: calibrate "wet" as the pot's soil at field capacity (soaked, drained 24 h) and "dry" at wilting, so sensor % ≈ available-water %
2	**Start window vs calendar.** Firmware and handoff: one run per day inside a start window, retry until the window ends. UI v2: any number of timed entries, retry until `qto`	Calendar wins; the window becomes "start time + up to `qto`". Delete `runS/startMin/endMin`
3	**Rain threshold.** The SA-28 is wet/dry only, so "Rain stop from 2 mm" can only use the forecast	Sensor wet → skip; Open-Meteo rain today ≥ `rainmm` → skip. Mock status skips closed-loop zones on rain but its "next" prediction does not: make both skip only non-closed-loop zones
4	**Daily limit.** Mock refuses when today + sec > cap; `WaterController` trims the run to what is left	Trim, refuse only when less than 60 s is left. Persist seconds today across reboots
5	**Pump alone 10 s against closed valves.** A dead-headed submersible pump heats and builds pressure; the valves are rated 0.6 bar	Check the pump datasheet; if unsure, drop that option and always open a valve with the pump
6	Frost default: `Config.h` 0 °C, handoff 4 °C, mock 2 °C	One value in `Config.h`; the mock copies it
7	Covered: handoff and firmware have an "until" date; UI v2 has a switch	Switch only
8	Button: "stop at once" (press) vs "on release" (deficit); does `btn = 0` also disable the All-off toggle; long press while All off is on is unspecified	As in 7.8: stop and All-off toggle at press-down, `btn = 0` disables only the deficit start (stop, Winter off and mute always work). Rest decided 30 Oct, see 7.8
9	Buzzer test: mock 1 s, API doc "two beeps"	Two beeps

#	Issue found	Proposal
10	History reads the whole `/api/log`; firmware v1 serves only the last 6 KB	Stream the last 500 rows (about 30 KB) with chunked transfer; full file stays at `/log.csv`
11	Reboot during a run writes no log row	Keep the active run in `RTC_NOINIT` memory; on boot write its row with result `FAULT` and the seconds it got
12	Closed/open loop "soil above limit" skip has no UI word	Add `r_WET` ("soil moist enough" / "elég nedves") to both languages
13	Hostname change needs mDNS restart and a DHCP reconnect	Restart mDNS at once; the router sees the new name at the next reconnect
14	DST: a 02:30 weekly entry does not exist on the spring-forward day and occurs twice in autumn	Fire by local date + minute, once per date (fired-tracking handles autumn; spring skips silently)
15	No login on the API; `/api/wifi/reset` is reachable by anyone on the LAN	Accept for the TDK; state it as a limit
16	`clog` bit 8 (season) is unused	Confirm seasons are never logged as COST rows

Implementation order

The experiment needs Schedule and Closed loop with the v2 log by about 3 November to give the paper (11 November) a week of data; everything else can land after it starts.

1. **Decide** items 1–5 of section 9 and freeze the v2 log format. (30–31 Oct)
2. **Hardware layer:** calibration setters, daily cap from settings, `StopReason::DND`, pump seconds per run, persisted seconds today, Button `PRESS` + 3 s long press. (1 Nov)
3. **Storage and log:** `Settings` v2, `CalendarStore`, `RunLog` v2 with RAM caches for `last` and `hist`. (1–2 Nov)
4. **Core logic:** `Rules`, `DndManager`, `RunQueue`, `CalendarScheduler`, `Deficit`, `ModeEngine` (Schedule + Closed loop with fallback). (2–3 Nov)
5. **Api v2 core** with ArduinoJson: state, zone GET/POST, sched, calendar, dnd, dsch, water, stop, global, settings, cal, clock, faults. Build `AppPage.h` from the v2 page without the mock. (3 Nov)
6. **Bench checks, then start the experiment:** Water now precedence, tank-low stop, reboot mid-run, DND stopping a run, log rows. (3–4 Nov)
7. **AlertMonitor, CostMeter**, `/api/cost/reset`, `/api/log` streaming. (4–6 Nov)

8. **WeatherClient task, WaterBalance, Open loop**, `/api/weather`; validate open loop in the software backtest, not on the model. (6–9 Nov)
9. **TestRunner** + `/api/test`, **ButtonActions + Feedback** patterns, `/api/wifi/reset`, SNTP. (9–13 Nov)
10. **Cleanup for the demo:** code-review items D5, D8, D11, C2, C3; host tests for `clamp()`, the deficit formula and DND windows across midnight. (14–17 Nov)